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Knowledge Fusion in Deep Learning-based Medical Vision-Language Models: A Review

Dexuan Xu[a], Yanyuan Chen[b], Zhongyan Chai[b], Yifan Xiao[b], Yandong Yan[a], Weiping Ding[d], Hanpin Wang[a], Zhi Jin[a], Wenpin Jiao[a], Weihua Yue[f], Hang Li[e], Yu Huang[c,*]

[a]*Key Laboratory of High Confidence Software Technologies, School of Computer Science, Peking University, Beijing 100871, China*
[b]*School of Software and Microelectronics, Peking University, Beijing 100871, China*
[c]*National Engineering Research Center for Software Engineering, Peking University, Beijing 100871, China*
[d]*School of Information Science and Technology, Nantong University, Nantong 226019, China*
[e]*the Department of Dermatology, Peking University First Hospital, Beijing 100034, China*
[f]*Peking University Sixth Hospital, Beijing 100191, China*

Abstract

Medical vision-language models based on deep learning can automatically extract image features and fuse them with text information, which has promoted the rapid development of multimodal medical artificial intelligence. However, the complexity of the medical field requires the model to have a deep professional knowledge background. Therefore, knowledge fusion technology provides a new idea for solving medical vision-language tasks. Different from the existing reviews, this paper systematically sorts out the knowledge fusion methods in medical vision-language models from two unique perspectives: the stage characteristics of knowledge fusion and the task-oriented fusion strategy, and provides a new theoretical framework for research in the field. Firstly, this paper introduces the classification of medical knowledge and its applicable scenarios in detail. Subsequently, we systematically discuss the knowledge fusion algorithm based on deep learning and summarize the four different knowledge fusion stages (data construction, pretraining, feature representation and inference) in the medical vision-language model. In addition, this paper comprehensively analyzes the specific strategies of knowledge fusion in five types of medical vision-language tasks (medical report generation, medical visual question answering, medical language-guided segmentation, medical multimodal pretraining, and multimodal large language model), and summarizes the evaluation methods based on knowledge fusion in detail. Finally, we summarize future research directions, including enhanced interpretability, mixture-of-experts models, knowledge editing, etc., aiming to provide researchers with references that have both theoretical value and practical significance.

Keywords: Knowledge fusion, Medical vision-language model, Deep learning, Multimodal

1. Introduction

With the rapid development of artificial intelligence in medicine, medical vision-language tasks have gradually become the frontier of medical artificial intelligence research. These tasks aim to achieve deep integration and interaction of visual and language information by combining medical images (X-rays, CT, MRI, etc.) with clinical texts (medical records, diagnostic reports, etc.). Medical vision-language tasks cover a number of specific areas, including medical report generation [1], medical visual question answering [2], medical language-guided segmentation [3], medical multimodal pretraining [4], and medical multimodal large language models [5]. They are not only of great value in clinical diagnosis, but also provide new ideas and tools for medical resource optimization, medical education, and interdisciplinary research.

The rise of medical vision-language models is an important milestone in the field of multimodal artificial intelligence. This technology provides strong technical support for the realization of medical vision-language tasks [7]. Through deep learning, medical vision-language models can automatically extract visual features from medical images, and associate and fuse them with text information, thereby realizing efficient processing of multimodal data. Specifically, the medical report generation model can generate relevant reports based on given medical images; the medical visual question-answering model can understand natural language questions and provide accurate answers based on the image content; the medical language-guided segmentation model supports the detection and segmentation of key parts in the image based on user-input instructions. In addition, vision-language pretraining models and medical multimodal large language models are also important research directions, laying the foundation for more general solutions to medical multimodal tasks.

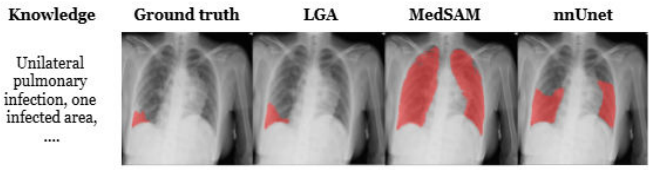


Figure 1: Comparing various model outputs to the ground truth mask. The knowledge fusion-based approach (LGA [6]) achieves closer alignment with the reference, highlighting the role of knowledge fusion in enhancing precision for critical clinical tasks.

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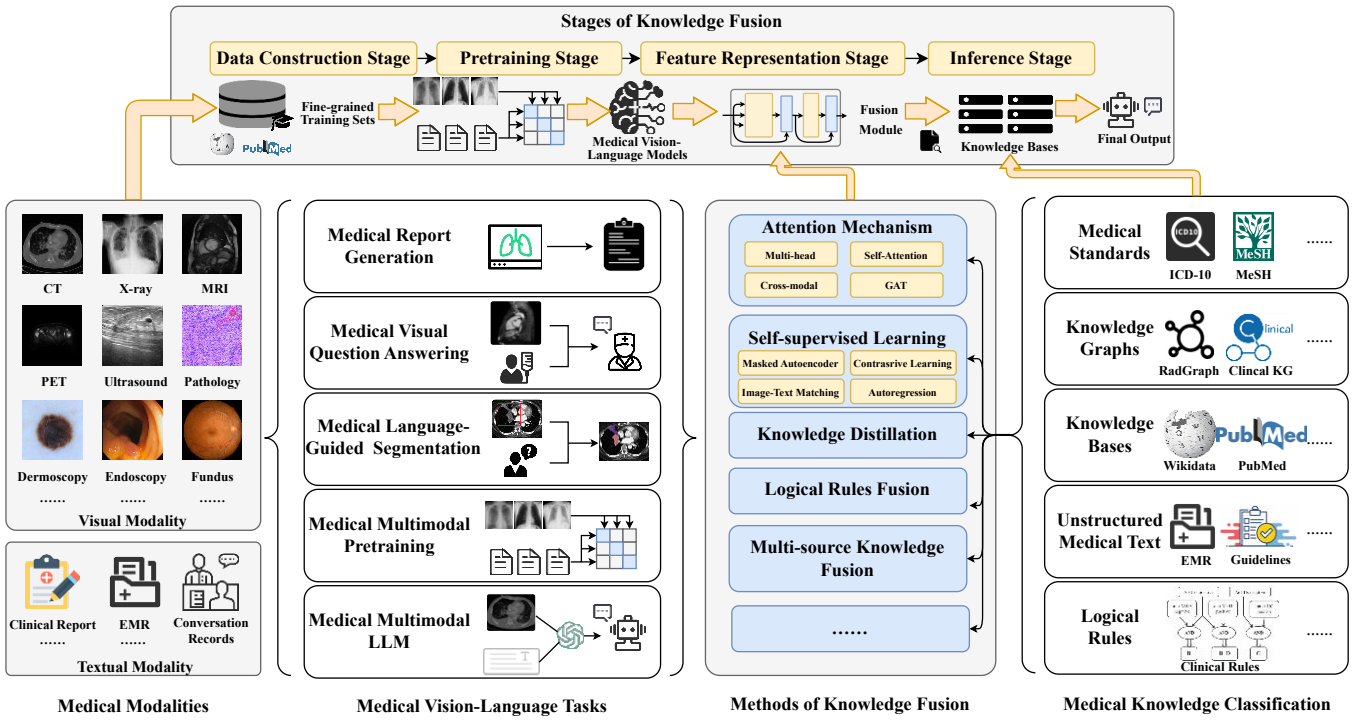


Figure 2: The overview of knowledge fusion in medical vision-language deep learning models. This review mainly covers five categories of medical vision-language tasks, including medical report generation, medical visual question answering, medical language-guided segmentation, medical multimodal pretraining, and medical multimodal large language model.

However, medical vision-language models also face many challenges in practice, especially in the deep integration with medical knowledge [7]. The complexity and professionalism of the medical field require that the model not only accurately understand the visual information in medical images, but also have a deep medical knowledge background to ensure that the generated text information is accurate, reliable and consistent with clinical reality. For example, in the task of generating medical reports, the model may generate descriptions that are inconsistent with medical facts due to lack of professional understanding [8]; in the task of medical visual question answering, the model has limited understanding of complex medical terms and it is difficult to provide accurate and clinically valuable answers [9]. In addition, the heterogeneity between medical images and clinical texts makes it difficult for the model to accurately match visual features with medical terms, especially when dealing with rare diseases or complex cases, the model's generalization ability is severely limited [10]. These challenges highlight the importance of knowledge in medical vision-language models.

Knowledge fusion is a specific information processing technology that aims to build a unified, consistent and semantically related knowledge system by integrating knowledge from different sources and in different forms [11]. It is not just a simple stacking of data or information, but through deep semantic understanding and logical reasoning, it integrates scattered knowledge fragments into a complete knowledge graph or knowledge base to provide support for the solution of complex problems. Compared with data fusion and information fusion, it not only integrates data and information, but also introduces domain knowledge, expert experience and logical reasoning. For medical vision-language models, knowledge fusion can align the visual information in the image with the pathological features in the medical knowledge base to improve the accuracy of diagnosis [12]. At the same time, knowledge fusion makes the decision-making process of the model more transparent and explainable by introducing medical knowledge [13]. To illustrate, medical decision-making is inherently high-risk and demands exceptional precision, yet traditional deep learning methods relying solely on visual data often fall short of this necessity. For example, for the segmentation tasks depicted in Figure 1, a system's ability to accurately delineate a lesion is significantly enhanced when it can fuse textual knowledge with visual cues [6]. Conversely, models lacking robust knowledge fusion may yield less precise segmentations. Such seemingly subtle inaccuracies like underestimating the extent of an infection or misidentifying its precise boundaries can have profound clinical ramifications, potentially leading to inaccurate prognostic assessments or delayed critical interventions.

At present, knowledge fusion has been widely used in medical vision-language models and covers all stages. First, when constructing training data, knowledge fusion at the data level can be performed [14–17]. Second, in the pretraining stage, targeted knowledge fusion pretraining can be implemented [10, 18–21]. In the feature fusion stage, feature-level knowledge fusion can be achieved through carefully designed knowledge extraction and fusion modules [9, 12, 22, 23]. Finally, external knowledge can also be fused during the infer-

ence process [24–27]. The schematic diagram of the knowledge fusion process is shown in Figure 2. Currently, the field is developing at a rapid pace, resulting in the emergence of different knowledge fusion strategies in various medical vision-language tasks, so it becomes crucial to provide researchers and practitioners with a comprehensive review. While several surveys have explored aspects of medical vision-language models [28, 29], a dedicated and comprehensive analysis of knowledge fusion across the diverse medical vision-language tasks remains less developed. Many existing works either focus on broader applications without a deep dive into knowledge integration strategies or cover specific tasks without a holistic view of how knowledge fusion is implemented and evaluated across the field. The purpose of this review article is to provide a comprehensive overview of the latest progress in knowledge fusion methods in deep learning-based medical vision-language models, and to provide future directions for knowledge fusion under the trend of large language models. The main contributions of this paper are as follows:

- This paper introduces the existing classification of medical knowledge in detail, and summarizes the applicable scenarios for each type of medical knowledge.
- This paper systematically discusses the knowledge fusion technology based on deep learning and summarizes the four stages of knowledge fusion in medical vision-language model.
- This paper comprehensively introduces how current work fuses knowledge in five types of medical vision-language tasks, including medical report generation, medical visual question answering, medical language-guided segmentation, medical multimodal pretraining, and multimodal large language models. At the same time, this paper also summarizes the current knowledge evaluation methods.
- This paper summarizes the challenges that knowledge fusion faces in medical vision-language models and gives suggestions on future directions.

The rest of this paper is organized as follows. Section 2 details the current classification of medical knowledge. Section 3 discusses the knowledge fusion techniques based on deep learning. Section 4 presents the four stages of knowledge fusion in the medical vision-language model. Section 5 details how existing works perform knowledge fusion in five categories of medical vision-language tasks. Section 6 introduces the evaluation method for the knowledgeability of the generated results. The challenges and future directions are introduced in Section 7, followed by the conclusion in Section 8. The organization layout of this review is shown in Figure 3.

2. Overview of Medical Knowledge Classification

Medical knowledge is the basis for medical vision language models to realize knowledge fusion, and the degree of its diversity and richness directly affects the basic capability and interpretability of the models. This section systematically describes the classification of knowledge types in the medical domain, including the categories of medical standards, knowledge graphs, specialized knowledge bases, Internet information bases, and unstructured medical text knowledge, as well as the characteristics of each category. Summarization of each knowledge is listed in Table 1.

2.1. Medical Standards

Medical standards are a series of systems for standardization and structured coding of clinical diagnosis, disease classification and medical subject terms. Its core value is to provide a unified semantic framework and terminology system for medical research, clinical management and information retrieval. When the model annotates, extracts or reasons about the information, the unified standard system is conducive to the elimination of ambiguities, so as to enhance the efficiency and precision of cross-system and multi-source knowledge fusion.

International Classification of Diseases (ICD) [30]. The ICD Classification List, developed by the World Health Organization (WHO), is a globally recognized standard for coding and classifying diseases, conditions and medical procedures. It provides a detailed framework for the systematic organization and reporting of health information in different clinical settings and countries, and serves epidemiological research, health management and clinical applications. The tenth version of ICD, i.e. ICD-10, contains a total of 12,119 disease entities, 3,228 symptomatic entities, and 7,504 drug entities. An illustration of ICD-10 is shown in Figure 4.

Medical Subject Headings (MeSH) [31]. The Medical Subject Headings (MeSH) thesaurus is a controlled vocabulary and is used for indexing, cataloging, and searching biomedical and health information and documents. The MeSH provides a consistent way of locating content that is terminologically different but conceptually the same. The MeSH organizes its descriptors in a hierarchical structure organization of its descriptors so that broad searches can find more narrowly indexed articles. This structure also provides an efficient way for searchers to navigate the grid to find the right descriptor. The MeSH glossary is continually updated by subject matter experts in a variety of fields. Many synonyms, near-synonyms, and closely related concepts are included in the entries to help users find the most relevant MeSH descriptor for the concept they are looking for.

International Classification of Functioning, Disability and Health (ICF) [32]. The International Classification of Functioning, Disability and Health (ICF) standardizes information on how disability, health, and environment affect a person's ability to perform usual activities (function). It is a standardized structure, or framework, for describing functioning and disability, which is applicable to all people, regardless of health condition.

Anatomical Therapeutic Chemical (ATC) [33]. ATC stands for Anatomical Therapeutic Chemical. The ATC code is the official classification system of the WHO for medicines, and is used as an international classification system for medicines. The ATC system categorizes drugs into five classes, consisting of letters spaced by numbers, with a total of seven digits.

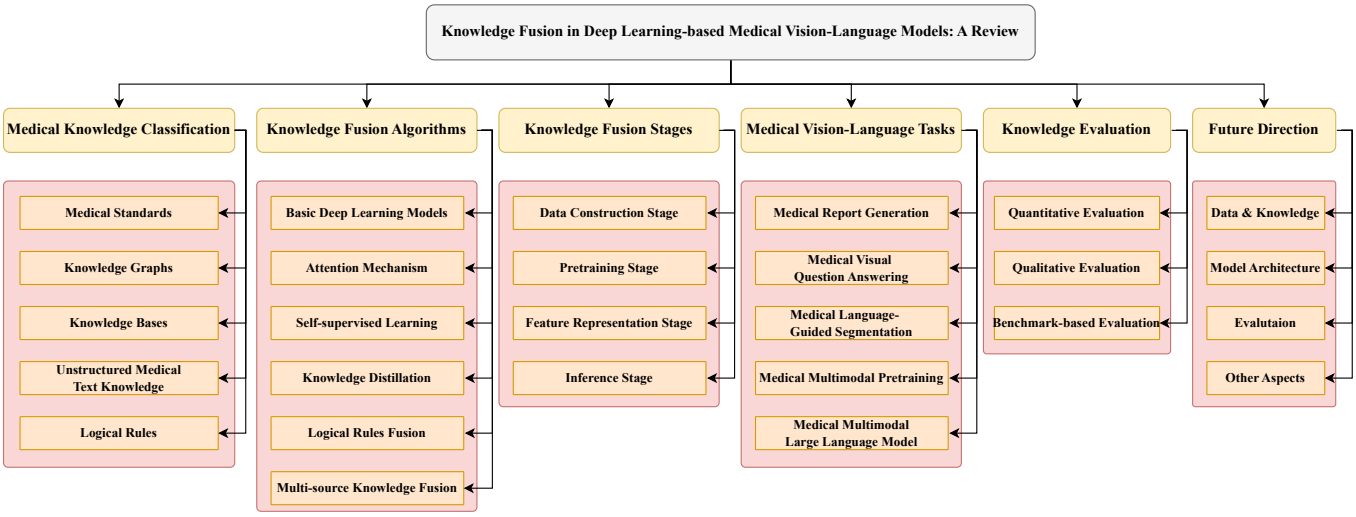


Figure 3: The organization layout of this review.

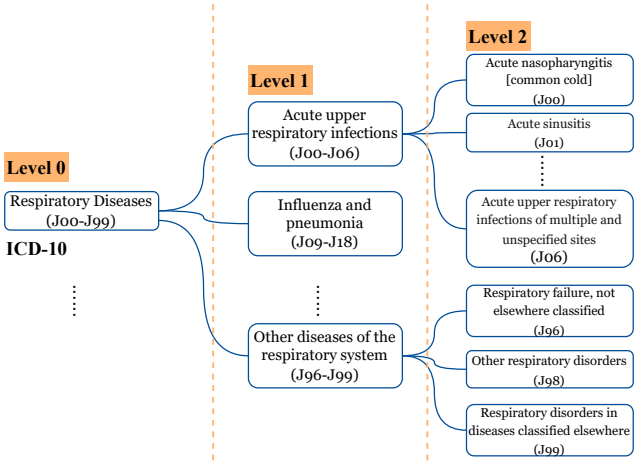


Figure 4: An illustration of ICD-10 medcial standard.

Medical Dictionary for Regulatory Activities (MedDRA) [34]. Medical Dictionary for Regulatory Activities is a collection of medical standardized terminology. MedDRA can be used for signal detection and clinical symptom monitoring for symptoms that include multiple systems or organs. MedDRA contains approximately 80,000 terms, each with a unique 8-digit arabic numeral code, at five levels: Low Level Terminology (LLT), Preferred Terminology (PT), High Level Terminology (HLT), Higher Level Group Terminology (HLGT), and System and Organ Classification (SOC).

2.2. *Knowledge Graphs*

A knowledge graph is a commonly used structured semantic knowledge base that can describe real-world concepts and relationships in symbolic form, with the basic unit of composition being the < *Entity* – *Relationship* – *Entity* > triples. Each entity can contain key-value pairs of attributes related to it, constituting a complete network structure. The medical knowledge graph provides interpretable reasoning paths and comprehensive knowledge support for medical vision-language modeling by integrating multi-source data.

2.2.1. *Unimodal Knowledge Graph*

LMKG. LMKG [35], is a large-scale, high-quality, multi-source, and multi-lingual medical knowledge graph, including 13 entity types and 17 relation types, containing 403,784 entity and 1,225,097 relation instances. Knowledge source information is also retained in the knowledge graph to preserve the variability of different information sources.

CMeKG. CMeKG[1] uses natural language processing and text mining technology to develop chinese medical knowledge graph based on large-scale medical text. It is constructed with reference to authorized international medical standard including ICD, ATC, SNOMED, MeSH, etc. and different clinical guide, industrial standard, diagnostic regulation and medical encyclopedia. CMeKG covers 11,076 kinds of diseases, 18,471 types of drugs, 14,794 types of symptoms and 3,546 kinds of clinical technology. It also has a structured description, which includes clinical symptoms, sick parts, drug treatment, and surgery treatment. The structured description further involves identification and diagnosis, imaging checks, high-risk factors, and spread channels. It also covers disease-prone populations, clinical departments, drug composition, and symptoms. Moreover, it includes details such as usage and dosage, validity, and contraindication. In total, there are over 30 kinds of common relationship types in the structured description of CMeKG. There are over 1.5 million types of concept relationship example and attribute triad.

CPubMed-KG. CPubMed-KG[2] is a large-scale open Chinese medical knowledge graph, which is mined from more than 2 million high-quality chinese core medical journals under the Chinese Medical Association, covering 534,842 diseases, 184,253 kinds of drugs, 142,415 kinds of symptoms, and all kinds of diagnostic and treatment techniques, totaling about

[1] https://cmekg.pcl.ac.cn/
[2] https://cpubmed.openi.org.cn/graph/wiki

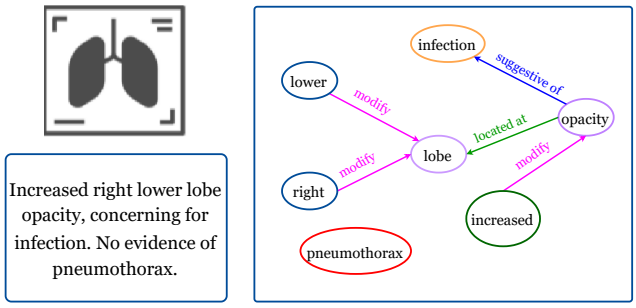


Figure 5: An illustration of RadGraph.

1,720,000 entities; it covers more than 40 kinds of relationships, such as medication, laboratory tests, etc.

Disease-KG. The main knowledge graph in Disease-KG [3] consists of 8 types of entities (Disease, Drug, Food, Check, Department, Producer, Symptom, and Cure) totaling 44,656 pieces of data, respectively; and 11 types of relationships totaling 312,159 pieces of data, where the *Disease* entity has additional attributes compared to other entities .

Clinical Knowledge Graph (CKG). CKG [36] is an open source platform currently comprised of more than 16 million nodes and 220 million relationships to represent relevant experimental data, public databases and the literature. The CKG also incorporates the latest statistical and machine learning algorithms, drastically accelerating analysis and interpretation of typical proteomics workflows. CKG integrates knowledge from multiple widely-used biomedical databases, serving as a foundational graph database.

2.2.2. Multimodal Knowledge Graph

In recent years, some work has begun to build multimodal knowledge graphs. Multimodal knowledge graphs contain knowledge of other modalities, such as images, in addition to textual modal information.

RadGraph. RadGraph [37] is an entity and relationship dataset containing full text radiology reports. It uses the raw medical report data MIMIC-CXR [38] and CheXpert [39] to construct the entity and relationship dataset, where annotations can be mapped to the relevant chest radiographs in the MIMIC-CXR and CheXpert datasets, respectively. The dataset consists of reports, entities, and relationships, which are de-identified in accordance with the U.S. Health Insurance Portability Act. The dataset is intended to support the development of natural language processing methods for radiology entity and relationship extraction, and to support multi-modal use cases that can utilize entities, relationships, and associated radiographs. An illustration of RadGraph is shown in Figure 5.

PrimeKG. PrimeKG [40] integrates 20 high-quality biomedical resources describing 17,080 diseases with 4,050,249 relationships, representing ten major biological scales. Textual descriptions of drug and disease clinical guidelines are added to PrimeKG's graph structure for multimodal analysis.

[3] https://github.com/nuolade/disease-kb

2.3. Knowledge Bases

Medical knowledge bases cover structured or semi-structured summaries of scientific literature, public databases, and clinical data. They are the main source of medical knowledge, and also provide the raw data and factual basis for the subsequent construction of medical vision-language models.

2.3.1. Specialized Medical Knowledge Base

PubMed. PubMed [4] is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health globally and personally. The PubMed database contains more than 37 million citations and abstracts of biomedical literature. It does not include full text journal articles; however, links to the full text are often present when available from other sources, such as the publisher's website or PubMed Central (PMC).

Unified Medical Language System (UMLS). The UMLS [5] is a health and biomedical vocabulary. It provides a comprehensive standardized list of medical terms, enabling data extraction in a structured way. UMLS consists of three main knowledge sources: the Metathesaurus, the Semantic Network, and the SPECIALIST Lexicon and Lexical Tools. The Metathesaurus is the core component of the UMLS, which includes 14 million entity names collected from 209 different terms in 25 languages. The Metathesaurus links entity names to different source terms that may have different formats but share the same or similar concepts, and therefore maps them to the same CUI. With over 3.8 million concepts and 7.8 million relationships in Metathesaurus, extensive medical knowledge coverage is available. An illustration of UMLS is shown in Figure 6.

2.3.2. Internet Information Repository

Wikipedia. Wikipedia [6] is a free Internet-based encyclopedia, started in 2001, that operates under open source management and allows users to edit almost any article. Wikipedia is the largest and most popular general reference on the Internet.

Medical Baike. Medical Baike [7] is a Chinese language knowledge base, similar to Wikipedia, that can be searched for relevant medical knowledge and edited by the user.

WikiData. WikiData [8], a free and open knowledge base with 115,665,464 data items that can be read and edited by both humans and machines. It encompasses structured data from various sources such as Wikipedia, Wikimedia Commons, and other wikis associated with the Wikimedia movement.

2.4. Unstructured Medical Text Knowledge

Electronic health records (EHR) contain a large amount of semi-structured information, recording patient history, symptoms, complaints, examination results, diagnostic results, etc., which is a real data source of medical diagnosis and treatment

[4] https://pubmed.ncbi.nlm.nih.gov/
[5] https://www.nlm.nih.gov/research/umls/index.html
[6] https://www.wikipedia.org/
[7] https://www.yixue.com/
[8] https://www.wikidata.org/

Table 1: The classification of medical knowledge sources.

Category	Subcategory/Name	Description	Key Data
Medical Standards	ICD-10	International Classification of Diseases (WHO standard)	12,119 disease entities, 3,228 symptom entities, 7,504 drug entities e.g. *Acute sinusitis(J01), Acute upper respiratory infections(J00-J06)*
	MeSH	Medical Subject Headings (biomedical thesaurus)	Hierarchical terminology with continuous expert updates e.g. *(MeSH Heading: Hypertension; Tree Number(s): C14.907.489; ...)*
	ICF	International Classification of Functioning framework	Standardized disability/function assessment system e.g. *Global mental functions (b110-b139), Consciousness functions(b110)*
	ATC	Anatomical Therapeutic Chemical classification	5-level drug coding system (7-digit codes) e.g. *Alimentary tract and metabolism(A), Metformin(A10BA02)*
	MedDRA	Medical Dictionary for Regulatory Activities	80,000 terms, 5-level structure (LLT/PT/HLT/HLGT/SOC) e.g. *(PT, Anaemia, 10002034)*
Knowledge Graphs	*Unimodal Knowledge Graphs*		
	LMKG	Multilingual medical knowledge graph	13 entity types, 17 relation types (403K entities, 1.2M relations) e.g. *(Distomolar<Body Structure>, Part of, Entire face<Body Structure>)*
	CMeKG	Chinese Medical Knowledge Graph	11K diseases, 18K drugs, 1.5M triples with 30+ relation types e.g. *(COVID-19, clinical manifestations, fever)*
	CPubMed-KG	Chinese medical journal KG	1.72M entities, 4.38M triples (40+ relation types) e.g. *(pelvic fracture, pathological classification, transverse fracture)*
	Disease-KG	Disease-centered knowledge graph	8 entity types (44K), 11 relation types (312K) e.g. *(Austin-type infantile metachromatic leukodystrophy, diet, duck meat)*
	CKG	Clinical Knowledge Graph	16M nodes, 220M relationships e.g. *(ERBB2<protein>, ASSOCIATED WITH, breast cancer<disease>)*
	Multimodal Knowledge Graphs		
	RadGraph	Radiology report graph with image links	Built from MIMIC-CXR & CheXpert datasets e.g. *([image-report pair], pneumothorax, OBS-DP)*
	PrimeKG	Multiscale biomedical KG	17K diseases, 4M relationships with clinical guidelines e.g. *(Autism, associated, DRD1, [text of Autism: ...])*
Knowledge Bases	*Specialized Bases*		
	PubMed	Biomedical literature database	37M+ citations with links to full texts e.g. *(PMID:29565029, title: Hypertension, abstract:....)*
	UMLS	Unified Medical Language System	14M entity names, 3.8M concepts (209 sources) *e.g. (Addison Disease, Mesh, D000224)*
	Internet Repositories		
	Wikipedia	Open encyclopedia	Largest general reference resource
	Medical Baike	Chinese medical wiki	User-edited medical knowledge base
	Wikidata	Structured open knowledge base	115M+ data items from Wikimedia
Unstructured Text	EHRs	Electronic Health Records	Semi-structured clinical narratives
	Literature/Guidelines/Text Books	Expert knowledge sources	Peer-reviewed articles & clinical protocols
Logical Rules	Derived from Expert Experience	Encoded diagnostic/treatment rules	e.g. $\exists(tumor) \wedge \exists(wall) \wedge in(tumor,wall) \rightarrow TumorStage(T_2)$; $\neg\exists(bladder_wall) \rightarrow InvalidDiagnosis$; $nearby(tumor,ureter) \vee in(tumor,ureter) \rightarrow HydronephrosisRisk(+)$; $< p_1, p_2, p_3 >=< \exists(tumor),in(wall,box),CT(+)> \rightarrow SurgicalIndication.$

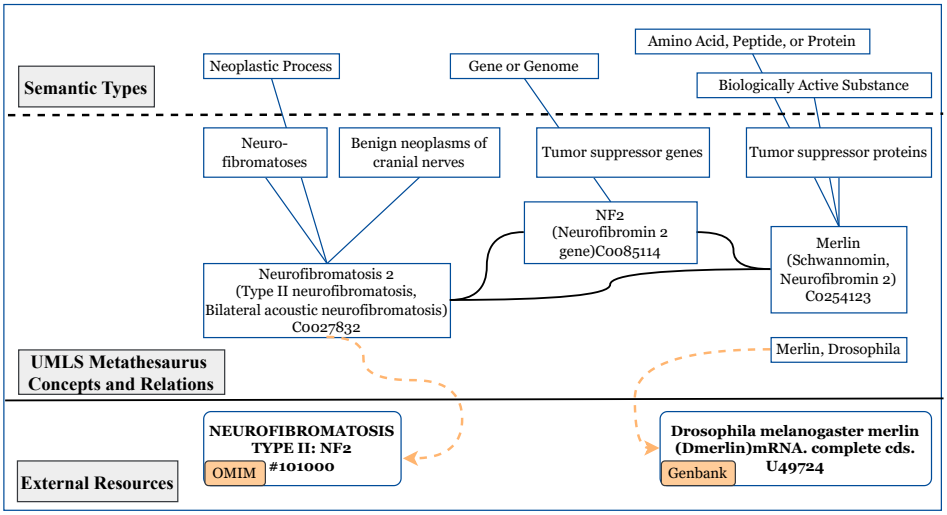


Figure 6: An illustration of Unified Medical Language System (UMLS).

knowledge. EHR data can provide real clinical data support for medical knowledge mapping (e.g., MIMIC-III/IV [41, 42], eICU [43], EHRSHOT [44],AmsterdamUMCdb [45]). In addition, a large number of guidelines, medical literature, and textbooks contain the most specialized expert knowledge and are important sources of medical knowledge.

2.5. Logical Rules

Medical guidelines, diagnostic procedures, treatment principles, etc., developed by clinical experts or professional groups can be abstracted into a series of logical rules or knowledge inference chains. By encoding these rules within the model or at the inference stage, the combination of logical rules helps to alleviate the network's dependence on data, allowing accurate results to be obtained even with only a limited number of labeled smaples [46].

3. Deep Learning Algorithms of Knowledge Fusion

Deep learning algorithms are an important foundation for knowledge fusion. In this section, we summarize the basic deep learning models for knowledge fusion in the medical field. We first introduce four types of basic deep learning models, and then introduce in detail the knowledge fusion algorithms based on attention mechanism, self-supervised learning, knowledge distillation, and logical rules fusion. The overview of these algorithms is shown in Figure 7.

3.1. Basic Deep Learning Models

3.1.1. Convolutional Neural Network

With the significant improvement of compute capability, Convolutional Neural Network (CNN), as an important algorithm in the field of deep learning, has achieved rapid development in recent years. Unlike the traditional method of manually extracting target features, convolutional neural networks can efficiently and automatically extract deep features in images, thereby significantly improving the accuracy of computer vision tasks such as image recognition and target classification [47], and showing extremely important value in medical multimodal tasks [48].

In 2012, the AlexNet [49] model won the image classification competition of the ImageNet Visual Recognition Challenge using convolutional neural networks. AlexNet contains 5 convolutional layers, 3 pooling layers, and 3 fully connected layers. In 2014, VGGNet [50] constructed a deep network by stacking 3×3 small convolution kernels, verifying the key role of network depth in feature abstraction capabilities. Similarly, GoogLeNet [51] introduced the Inception module, using multi-scale parallel convolution and 1×1 dimensionality reduction convolution to achieve higher classification accuracy on ImageNet with fewer parameters. In response to the gradient vanishing problem in deep network training, He et al. [52] proposed the ResNet network. The model constructs a jump path through residual connections, allowing the gradient to be directly back-propagated to the shallow layer, successfully training a 152-layer network. The residual learning paradigm inspired derivative architectures such as DenseNet [53], which further enhanced the feature reuse efficiency through dense cross-layer connections.

In the field of medical imaging, the innovative application of CNN continues to deepen. U-Net [54] achieves medical image segmentation through a symmetric encoding-decoding structure and jump connections. The encoder part gradually extracts high-level semantic features through 5 layers of convolution and downsampling. Each convolution layer contains two 3×3 convolution kernels, and the number of channels doubles from 64 to 1024 layer by layer. The decoder part gradually restores the spatial resolution through transposed convolution, and each layer fuses features with the corresponding layer of the encoder through jump connections to retain low-level detail information. U-Net and its derivative models are widely used in medical image segmentation tasks. In addition, medical images (such as CT, MRI) are usually high-dimensional and contain rich spatial or temporal information. 3D CNN [55] can process information in three-dimensional space more efficiently, extract

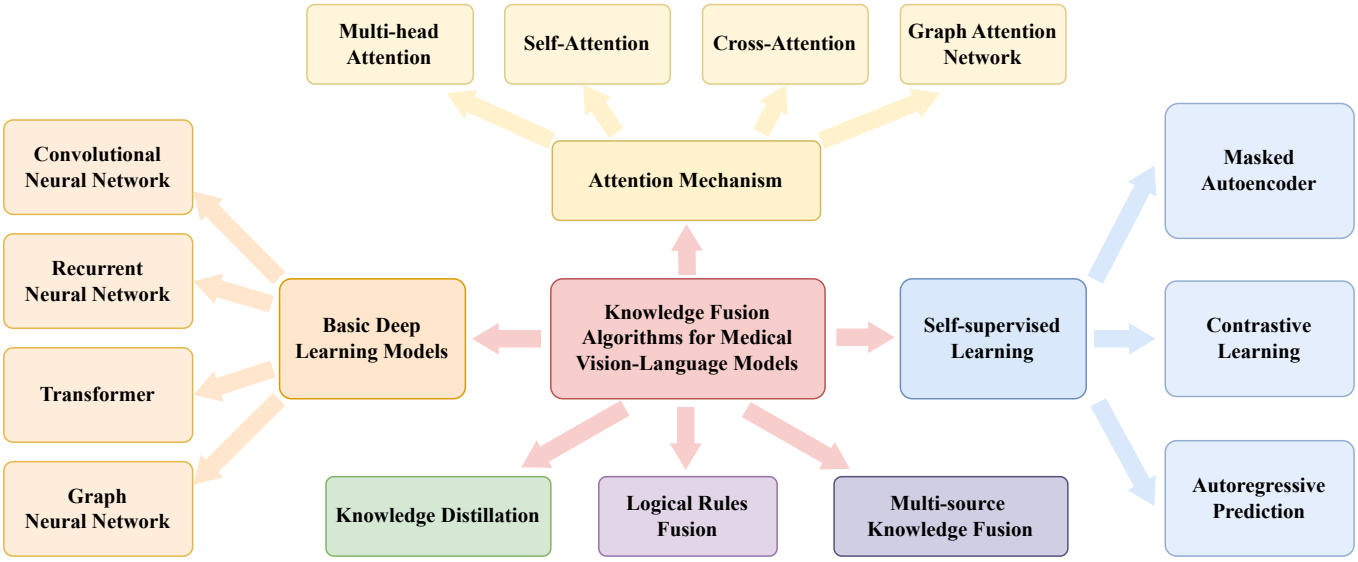


Figure 7: The overview of knowledge fusion algorithms for medical vision-language models.

more comprehensive features through three-dimensional convolution kernels, and better capture the spatial relationship between tissues. This model has become the research basis of high-dimensional medical image processing [56].

3.1.2. Recurrent Neural Network

Recurrent Neural Network (RNN) [57] is a type of deep learning model specifically designed for processing sequence data. RNN passes information between time steps through hidden states, allowing the model to capture the temporal dependencies of input data. This feature gives RNN a significant advantage in tasks that require contextual information. However, standard RNNs suffer from the vanishing and exploding gradient problems when processing long sequences of data, which limits their ability to model long-term dependencies.

As RNN research deepens, researchers have made many improvements to its structure. The most representative ones are Long Short-Term Memory (LSTM) [58] and Gated Recurrent Unit (GRU) [59]. The former realizes the effective storage and discarding of historical information through the design of input gate, forget gate and output gate, thereby alleviating the gradient vanishing problem and enabling the model to capture dependencies over a longer time span. The latter further simplifies the gate structure based on LSTM, using only update gate and reset gate, while improving computational efficiency and still maintaining strong sequence modeling capabilities.

In the medical field, recurrent neural networks and their variants are widely used in tasks such as biosignal analysis, electronic health record modeling, and medical text processing. In physiological signal analysis, LSTM is used for anomaly detection of electrocardiogram and electroencephalogram signals, which can effectively capture time-dependent patterns such as arrhythmias or epileptic seizures. In medical text processing tasks, RNN is used for medical named entity recognition [60], which improves the efficiency of medical literature analysis and intelligent diagnosis systems. In recent years, RNN and CNN have been widely combined and applied to various medical multimodal tasks, such as report generation [61, 62] and visual question answering [2].

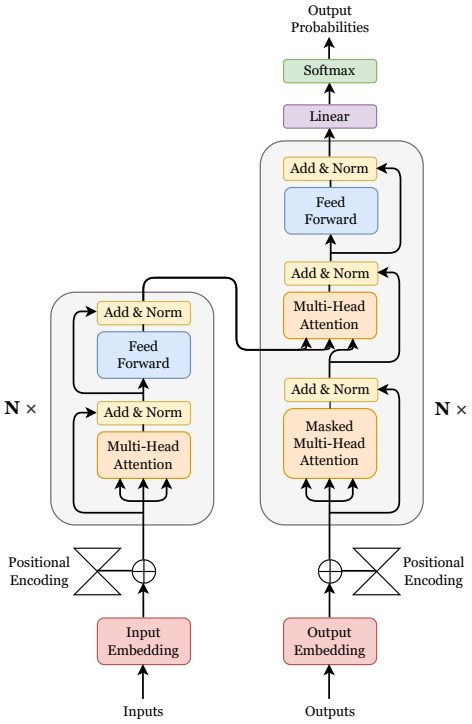


Figure 8: The architecture of Transformer [63].

3.1.3. Transformer

Transformer [63] is a deep learning model based on the self-attention mechanism. The architecture of Transformer is shown in Figure 8. After it was proposed, it quickly became the mainstream architecture in the field of natural language processing. Unlike traditional RNN, Transformer uses the self-attention mechanism to directly model global dependencies, thereby greatly improving the efficiency of parallel computing. This feature makes Transformer perform well in tasks such as

machine translation and text generation, and has become the basic architecture of current large-scale pretrained language models.

Researchers have optimized Transformer in many directions. In the field of natural language processing, BERT [64] improves text comprehension through bidirectional pretraining, and GPT [65] enhances text generation through autoregression. At the same time, Transformer is also developing in the field of computer vision. In 2020, Google proposed Vision Transformer (ViT) [66], which proved the feasibility of pure Transformer structure in image classification tasks for the first time and outperformed traditional CNN on large-scale datasets. Subsequently, Swin Transformer [67] further improved the performance of Transformer in dense vision tasks through a hierarchical window attention mechanism. Xiao et al. [68] proposed a Top-k Token Selective Transformer for remote sensing image super-resolution, designed a residual token selection group to dynamically select top-k keys for each query to capture key tokens. A multi-scale feed-forward layer was developed to generate multi-scale feature mixtures, and a global contextual attention mechanism was proposed to explore global contextual features with large receptive fields to improve remote sensing image super-resolution tasks.

In the medical field, Transformer is widely used in tasks such as medical text understanding and medical image analysis. In medical text processing, BioBERT [69] and ClinicalBERT [70] are pretrained specifically for electronic health records and biomedical literature, which greatly improves the performance of tasks such as medical named entity recognition, relationship extraction, and medical question answering. In medical image analysis, ViT is used for medical image classification, segmentation, and detection. TransUNet [71] combines Transformer and U-Net structures to show superior performance in medical image segmentation tasks. Wang et al. [72] proposed TransBTS, which combines 3D CNN with Transformer. For the 3D multimodal brain tumor segmentation task, 3D CNN is used to extract local spatial features, and Transformer is used to encode global features, which effectively models local and global features in spatial and depth dimensions. Dalmaz et al. [73] proposed ResViT for multimodal medical image synthesis. It combines local accuracy and context sensitivity in the residual bottleneck, and uses the aggregated residual Transformer block to synergistically retain local and global context, and performs well on multimodal brain MRI and pelvic MRI-CT datasets. As the application of Transformer in multimodal learning, it has also been used on a large scale in the fields of medicine for knowledge fusion and cross-modal information integration [73, 74].

3.1.4. Graph Neural Network

Graph Neural Network (GNN) is a deep learning model specially designed for processing graph structured data. It is widely used in knowledge graph reasoning, bioinformatics and other tasks. Unlike traditional neural networks, GNN can directly process non-Euclidean data and model complex relationships in graphs through neighbor information aggregation mechanism. This feature enables GNN to perform well in tasks such as node classification, link prediction, and graph classification, and demonstrates strong modeling capabilities in application scenarios that require processing structured data.

In recent years, researchers have proposed a variety of models to improve the performance of GNN in different tasks. Among them, the Graph Convolutional Network (GCN) [75] implements spatial convolution by approximating the simplified form of spectral graph convolution, so that nodes can effectively propagate and update information in the graph structure. The core calculation formula of GCN is as follows:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right), \tag{1}$$

where $\tilde{A} = A + I$ is the adjacency matrix plus self-connection, $\tilde{D}$ is the degree matrix of $\tilde{A}$, $H^{(l)}$ represents the node features of the third layer, $W^{(l)}$ is the trainable weight matrix, and σ is the activation function. The architecture of GCN is shown in Figure 9. In addition to GCN, the Graph Attention Network (GAT) [76] uses the attention mechanism to assign different weights when neighbors are aggregated, allowing the model to learn the importance between nodes more flexibly.

In the medical field, GNN is widely used in protein interaction prediction and knowledge graph reasoning. In medical knowledge graph reasoning, GNN supports clinical decision-making and personalized diagnosis and treatment by mining the implicit relationship between diseases, symptoms and drugs [77]. In brain network analysis, GNN can model the functional connection network constructed by fMRI data for early classification of Alzheimer's disease [78]. Despite its great potential, GNN still faces challenges such as data heterogeneity, small sample learning and interpretability in medical applications. Its research in cross-modal medical data will be an important direction in the future.

3.2. Attention Mechanism

In the process of multimodal knowledge fusion, the model needs to extract key knowledge from data of multiple modalities. The attention mechanism is a technology in deep learning that imitates the human visual and cognitive system to focus on important information. It enables the model to automatically focus on the key parts of the input data by dynamically assigning weights. The calculation formula of the attention mechanism is as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}, \tag{2}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ represent the matrices of Query, Key, and Value respectively, and d_k is the dimension of $\mathbf{K}$. They represent different meanings in different tasks. For the self-attention mechanism [63], $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ are obtained by multiplying the same input sequence vector with three different trainable parameter matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$. For the cross-attention mechanism [79], $\mathbf{Q}$ and $\mathbf{KV}$ usually come from two different modal sequences. In multimodal medical tasks, the use of cross-attention can help the model determine which visual information is more relevant to textual information, thereby more effectively integrating knowledge [80, 81].

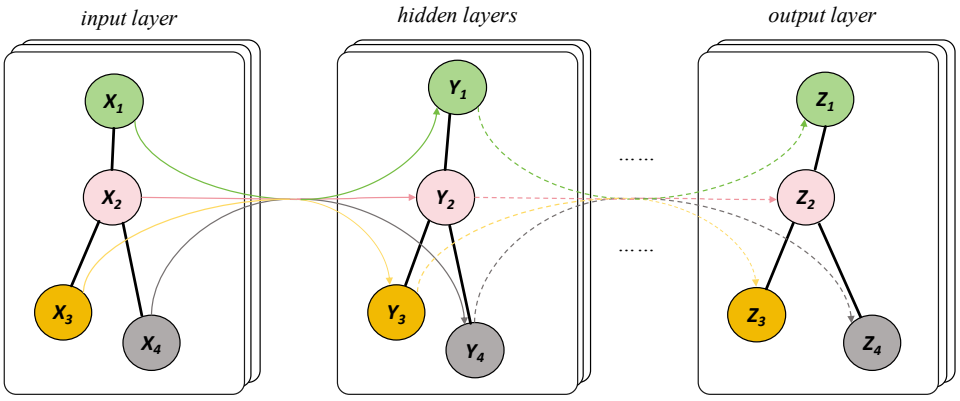


Figure 9: The architecture of Graph Convolutional Network [75].

3.2.1. Muti-head Attention

Furthermore, the multi-head attention mechanism is proposed:

$$A_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \text{softmax}\left(\frac{\mathbf{Q}_i\mathbf{K}_i^T}{\sqrt{d_{head}}}\right)\mathbf{V}_i, \quad (3)$$

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concact}\{A_i\}_{i=1}^{H}\mathbf{W}^O, \quad (4)$$

where Concact$\{\cdot\}$ is the concatenation operation, H is the number of heads, and $\mathbf{W}^O$ is a trainable matrix. Through the calculation of multiple groups of attention, the output of the attention layer can express the attention information in different subspaces, thereby enhancing the expressive power of the model. At present, multi-head attention is widely used in Transformer-based model architectures and has become a basic strategy for knowledge fusion at the feature level. Through various variants of multi-head attention, the model can better integrate information from different modalities, thereby improving the effect of various multimodal medical tasks [12, 22, 82, 83].

3.2.2. Graph Attention Network

For knowledge stored in graph structures, Graph Attention Network combined with attention mechanism is an important method for knowledge fusion. Specifically, for the knowledge graph $\mathcal{G}$, we define the attention A_{ij} between nodes N_i and N_j:

$$A_{ij} = \frac{\exp\left(\mathbf{a}^T\left[\mathbf{W}\mathbf{x}_i \| \mathbf{W}\mathbf{x}_j\right]\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(\mathbf{a}^T\left[\mathbf{W}\mathbf{x}_i \| \mathbf{W}\mathbf{x}_k\right]\right)}, \quad (5)$$

where $\mathbf{a}$ and $\mathbf{W}$ are learnable matrices, $\|$ represents feature concatenation, $\mathbf{x}_i$ is the input feature of node N_i, and $\mathcal{N}_i$ is the neighborhood set of node N_i. The output feature of each node can be calculated using multi-head attention:

$$\mathbf{x}_i^{'} = \mathbf{x}_i + \|_{h=1}^{H}\sigma\left(\sum_{k\in\mathcal{N}_i} A_{ik}^h\mathbf{W}^h\mathbf{x}_k\right), \quad (6)$$

where H represents the number of heads. Graph attention networks have been widely used in medical multimodal tasks combined with knowledge graphs [12, 81, 84] and have been shown to perform better than other graph learning methods.

3.2.3. Further Research on Attention Mechanisms

In the field of deep learning visual attention mechanisms, a variety of attention mechanisms have shown unique advantages [85]. Hou et al. [86] proposed the Cross Attention Network, which effectively enhances the overall discrimination ability of few-shot classification by calculating the correlation between query and target features. Huang et al. [87] proposed the criss-cross attention, which encodes the pixel context with an innovative cross-path, and has significantly improved the memory and time utilization efficiency compared with the traditional non-local block. Jaegle et al. [88] proposed the Perceiver, which uses iterative attention and Fourier transform to process multimodal data, and reduces the parameters to one tenth of the original through weight sharing, greatly enhancing the generalization ability of the model. Lee et al. [89] proposed stacked cross attention, which can effectively discover all possible alignment relationships between images and texts by calculating the cross attention between the two, significantly enriching the semantics of multimodal tasks.

In medical multimodal knowledge fusion, the attention weights calculated by the attention mechanism can be used to explain the decision-making process of the model [90]. Sun et al. [91] proposed a model based on mixed attention and CNN for endometrial tissue pathology image classification, and achieved excellent performance in related tasks. Zhao et al. [92] proposed a bilateral adaptive spatial and channel attention network to improve the network's feature extraction ability and explore the discriminative information of breast density classification. Woo et al. [93] proposed a convolutional block attention module, which calculates channel and spatial attention separately, achieving the effect of emphasizing useful channels and enhancing information-rich local areas in the image, reducing interference information. By analyzing the distribution of attention, these works can understand the focus of the model when fusing features, provide a basis for understanding the model behavior, and enhance the interpretability of the model.

3.3. Self-supervied Learning

Self-supervised learning is a learning method that performs supervised learning by utilizing the structure and characteristics of the data itself. This method does not rely on manually

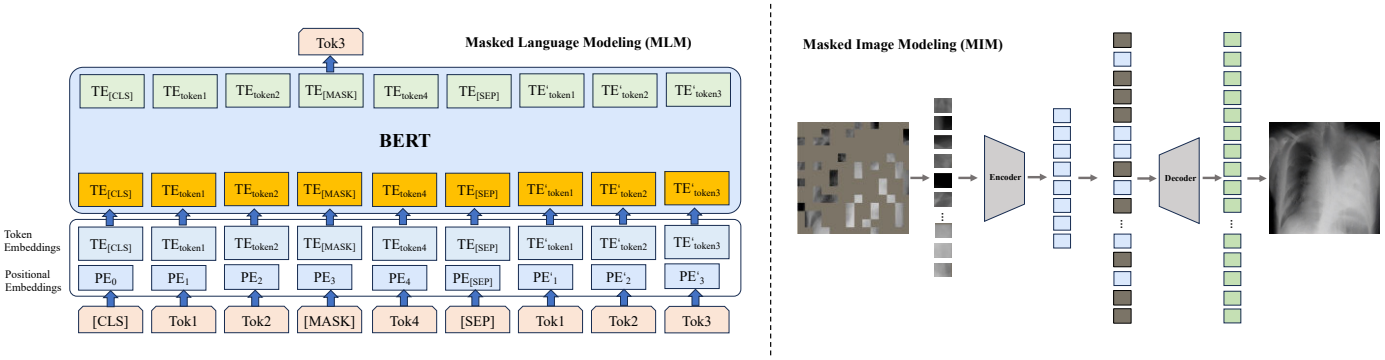


Figure 10: The architecture of Masked Language Modeling (MLM) and Masked Image Modeling (MIM).

labeled data, but automatically learns the inherent laws and feature representations of data from large-scale unlabeled data. For example, in natural language processing, the structure and semantics of language can be learned by predicting the missing words or the next word in a sentence.

Large-scale self-supervised learning provides a basis for knowledge fusion. Large-scale training data contains a large amount of information and knowledge, which can be captured and learned by self-supervised learning models. In multimodal learning, large-scale image and text data are used for pretraining to learn the association and correspondence between images and texts, thereby realizing the alignment and fusion of multimodal knowledge. In medical vision-language models, they mainly include methods based on masked autoencoders and methods based on contrastive learning.

3.3.1. Masked Autoencoder

Masked autoencoder (MAE) is an efficient representation learning model based on self-supervised learning. By randomly masking the input data, the model is forced to reconstruct the masked parts from the visible contextual information, thereby learning the intrinsic structure and robust representation of the data. MAE includes Masked Image Modeling (MIM) [94] and Masked Language Modeling (MLM) [64]. For MIM, given an input image $X \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height. width and the number of channels respectively, a random binary mask $\mathbf{M}_I$ is sampled under the control of the mask ratio hyperparameter γ. Each element $m_i \in 0, 1$ of the binary mask $\mathbf{M}_I$ corresponds to a pixel in the image. The masked image is defined as:

$$\hat{X}_m = X \odot (1 - \mathbf{M}_I), \tag{7}$$

where $\odot$ denotes the element-wise product. MAE framework consists of the encoder f_θ and the decoder g_ϕ parameterized by θ and ϕ, respectively, to reconstruct the masked image. In practice, the encoder f_θ is usually instantiated by Vision Transformers, where patch-wise masks are employed. The objective of MIM is formulated as follows:

$$\min_{\theta,\phi} \mathbb{E}[\mathcal{D}(g_\phi(f_\theta(\hat{X}_m) \odot \mathbf{M}_I), X \odot \mathbf{M}_I)], \tag{8}$$

where $\mathcal{D}(\cdot,\cdot)$ represents the distance metric to measure the reconstruction quality of the masked patches such as L1 loss.

Similarly, for MLM, we can obtain the masked text $\hat{T}_m$ by a random mask $\mathbf{M}_T$. The masked tokens are replaced with special character or another word according to the ratio τ. The objective of MLM is formulated as follows:

$$\min_{\theta,\phi} \mathbb{E}[C(g_\phi(f_\theta(\hat{T}_m) \odot \mathbf{M}_T), T \odot \mathbf{M}_T)], \tag{9}$$

where $C(\cdot,\cdot)$ represents the metric that measures the recovery accuracy of masked tokens, such as cross entropy loss. The architecture of MLM and MIM is shown in Figure 10.

For multimodal tasks, Li et al. [95] proposed SelfDoc, which predicts randomly masked language or visual features in document image understanding to help the model infer contextual clues and multimodal information. Sun et al. [96] proposed VideoBERT, which converts the raw input into a discrete token sequence and uses intra-modal masking to achieve the goal of pre-training on a video or text corpus. Bao et al. [97] proposed VL-BEiT, which uses a shared transformer to predict masks for unimodal and multimodal data, and can be learned from a unified pre-training task. In the medical field, masked autoencoder is an important method for learning knowledge from massive data. Zhou et al. [98] performed mask operations on radiograph-related records based on masked record modeling to learn image features and generate reports, thereby improving the accuracy and effectiveness of medical report generation. Georgesce et al. [99] used the masked autoencoder model to learn the structure of normal samples and trained the anomaly classifier by the difference between the original image and the reconstructed image, achieving excellent results in medical anomaly detection. Combined with the reconstruction task, the masked autoencoder can enhance the understanding of the image content and show unique advantages in medical multimodal image analysis, disease diagnosis and knowledge fusion [21, 100, 101].

3.3.2. Contrastive Learning

Contrastive learning is a self-supervised representation learning paradigm that learns discriminative features by distinguishing semantically similar samples (positive sample pairs) from dissimilar samples (negative sample pairs). Unlike masked autoencoders that rely on reconstruction tasks, the core goal of contrastive learning is to capture the high-level semantic invariance of data by narrowing the embedding distance of

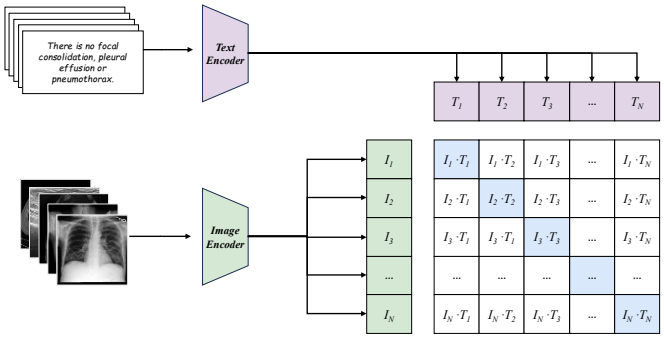


Figure 11: An illustration of multimodal contrastive learning methods (CLIP) [104].

positive sample pairs and pushing the embedding distance of negative sample pairs away.

Contrastive learning first requires augmenting the input data. For a given input sample, a positive sample pair is generated through a random augmentation strategy. For small batches of N samples, the InfoNCE loss [102] is usually used to optimize model:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\frac{\text{sim}(z_i,z_{i+})}{\tau}\right)}{\exp\left(\frac{\text{sim}(z_i,z_{i+})}{\tau}\right)+\sum_{j=1}^{K}\exp\left(\frac{\text{sim}(z_i,z_{i-j})}{\tau}\right)}, \quad (10)$$

where z_{i+} is the positive sample feature of z_i, z_{i-j} is the negative sample in the batch, $\text{sim}(\cdot)$ is the similarity calculation function, and τ is the temperature hyperparameter. Advanced designs often improve the diversity of negative samples through memory banks or momentum encoders. For example, the MoCo [103] framework maintains a dynamically updated queue of negative samples through a momentum encoder, achieving large-scale negative sampling without increasing computational costs. The illustration of multimodal contrastive learning [104] is shown in Figure 11.

Contrastive learning is widely used in data enhancement and multi-view data. Akbari et al. [105] proposed VATT using a similar contrast objective and studied a modality-independent single backbone network for multimodal self-supervised learning. Guzhov et al. [106] proposed AudioCLIP to extend CLIP to the fields of images, text and audio. Xiao et al. [107] proposed a self-supervised degradation guided adaptive network, which integrates contrastive learning into the blind remote sensing image super-resolution task. By using robust representation in the embedding space to characterize complex degradation, the downstream super-resolution network can combine degradation prior knowledge to improve adaptability and effectively deal with diverse and unknown image degradation situations in reality.

In the medical field, multimodal contrastive learning is widely used as a pretraining method to mine common knowledge between modalities from massive data and achieve multimodal knowledge fusion [19, 108]. Sowrirajan et al. [109] proposed MoCo-CXR, applying contrast learning to chest X-ray images to verify and expand its effectiveness and application scope in the medical field. Yang et al. [110] proposed a semi-supervised breast cancer histology classification method that combines deep multiple instance learning and contrast predictive coding. In the case of small samples, it can effectively improve the accuracy of breast cancer histology classification and reduce the dependence on a large number of labeled samples. These works improve the ability to understand and extract data features by allowing the model to learn to associate similar samples and distinguish different samples [20, 111].

3.3.3. Other Self-supervised Learning Methods

There are many other self-supervised learning strategies, such as multimodal matching, autoregressive prediction, etc., which are also widely used in pretraining to extract knowledge from pretraining data. The image-text matching (ITM) task was first proposed in UNITER [112], which inputs the global cross-modal representation into a binary classifier to predict whether the input image-text pair matches. ITM can also be used to complement contrastive learning. Currently, there are a lot of works [100, 113, 114] that set ITM as a basic self-supervised pretraining task.

Autoregressive prediction predicts the following result each time through the input of the previous context. SimVLM [115] improves autoregressive pretraining through the prefix language model objective and only performs autoregressive decomposition on the remaining tokens, effectively utilizing cross-modal information and reducing training complexity. UNIMO [116] uses masked language modeling and seq2seq autoregressive masked prediction to enhance the model's cross-modal understanding and generation capabilities. At present, autoregressive prediction has become a widely used pretraining strategy for large language models, and has been proven to significantly improve the model's language generation coherence and logical rationality [117]. In the field of medical diagnosis assistance [118, 119], with the continuous increase in model parameters and training data volume, the autoregressive prediction mechanism enables the model to not only remember the knowledge in the pre-training corpus, but also transfer and generalize knowledge, solving new problems in zero-shot or few-shot learning scenarios.

3.4. Knowledge Distillation

Knowledge distillation [120] is a model compression and acceleration technology. The core idea is to use a well-trained large model (teacher model) to guide the training of a small model (student model) so that the output of the student model is as close to the teacher model as possible. This method aims to train the student model through a combination of soft targets and hard targets, thereby achieving model compression and performance improvement. The learning objectives of knowledge distillation include soft target loss and hard target loss. Soft target loss uses Kullback-Leibler divergence to measure the difference between the probability distributions output by the student model and the teacher model. The formula is as follows:

$$\mathcal{L}_{soft} = \frac{1}{N}\sum_{i=1}^{N} KL\left(\frac{exp(z_i/\tau)}{\Sigma_j exp(z_j/\tau)}\|\frac{exp(\hat{z}_i/\tau)}{\Sigma_j exp(\hat{z}_j/\tau)}\right), \quad (11)$$

where z_i is the output of the teacher model, $\hat{z}_i$ is the output of the student model, and τ is the temperature hyperparameter. The hard target loss uses the cross entropy loss to measure the difference between the predicted output of the student model and the true label. The formula is as follows:

$$\mathcal{L}_{hard} = \frac{1}{N}\sum_{i=1}^{N} CE(y_i, \hat{y}_i), \tag{12}$$

where y_i is the true label, $\hat{y}_i$ is the prediction result of the student model. The total loss needs to be a weighted average of the soft target loss and the hard target loss:

$$\mathcal{L}_{total} = \lambda\mathcal{L}_{soft} + (1-\lambda)\mathcal{L}_{hard}. \tag{13}$$

In terms of medical knowledge fusion, knowledge distillation can integrate knowledge from different data sources or modalities to provide more comprehensive diagnostic information [121–123]. By distilling the knowledge of teacher models, the student model can learn richer feature representations and improve its generalization ability and robustness in practical applications.

3.5. Logical Rules Fusion

Logical rules provide a semantic and human-readable form of knowledge representation. Logic-based knowledge fusion methods [124] aim to improve the interpretability of models and reduce the reliance on large amounts of annotated data by combining logical rules with deep learning models. Usually, it is necessary to define atomic propositions and construct compound propositions. In medical scenarios, propositions can include existential propositions and spatial relationship propositions, etc., and then atomic propositions are combined through logical connectors to obtain compound propositions. In order to integrate prior logic into deep learning models, it is necessary to vectorize logical rules and obtain feature embedding. Specifically, for specific tasks, such as medical language-guided segmentation, the segmentation mask M is divided into regions, and existential propositions are generated, which are further converted into binary vectors through spatial relationship proposition geometry calculations. Afterwards, the logical rules are parsed into a tree structure, with leaf nodes as atomic propositions and internal nodes as logical operators. Finally, a graph neural network is used to map the logical rule tree to the latent feature space. During the training process, it is usually necessary to increase the consistency loss of logical rules:

$$\mathcal{L}_r = \sum_{i,j}\|R_{i,j} - \hat{R}_{i,j}\|_F^2 + \sum_i \|G_i - \hat{G}_i\|_F^2 + \sum_i \mathrm{BCE}(y_i, \hat{y}_i), \tag{14}$$

where $R_{i,j}$ and $\hat{R}_{i,j}$ are the atomic proposition embedding features of the truth value and prediction, G_i and $\hat{G}_i$ are the global rule embedding features of the truth value and prediction, and $\mathrm{BCE}(\cdot,\cdot)$ calculates the rule truth classification loss. This deep learning method that combines logical rules for knowledge fusion has been applied to bladder tumor segmentation [46].

Another logic-based knowledge fusion method is to combine fuzzy logic in the reasoning stage. Specifically, the input data needs to be converted into fuzzy sets, and the membership function is used to represent the degree of membership of the input data in different fuzzy sets. Then, reasoning is performed based on the fuzzy rules and the input data. At present, strategies combining fuzzy logic rules have been studied in various medical multimodal tasks [26, 27].

3.6. Multi-source Knowledge Fusion

In the context of big data, knowledge has the characteristics of multi-source heterogeneity, dynamic evolution, and semantic diversity [125]. How to effectively integrate fragmented knowledge into a structured knowledge base is the core challenge in the field of knowledge fusion. The goal of multi-source knowledge fusion is to integrate multi-source knowledge into a consistent and complete knowledge system through semantic alignment and association calculation. According to the technical path of multi-source knowledge fusion, this section divides it into Bayesian model fusion, evidence theory fusion, and emerging graph model fusion methods.

3.6.1. Fusion based on Bayesian Model

Since the quality of multi-source knowledge is affected by different data sources, the fusion method based on Bayesian model [126] uses prior probability and conditional probability to achieve consistent fusion of knowledge. Assume that the set of knowledge to be fused is $k = \{k_i | 1 \le i \le N\}$, and the multi-source observation is $S = \{S_j | 1 \le j \le M\}$, then the fusion posterior probability of knowledge k_i is:

$$P(k_i|S) = \frac{P(S|k_i)\cdot P(k_i)}{P(S)}, \tag{15}$$

where $P(S) = P(S_1, S_2, \ldots, S_M)$ is the joint probability distribution of the observation value. Assuming that $\{S_1, S_2, \ldots, S_M\}$ are independent of each other, we have:

$$P(k_i|S) = \frac{\prod_{j=1}^{M} P(S_j|k_i)\cdot P(k_i)}{\sum_{i=1}^{N}\prod_{j=1}^{M} P(S_j|k_i)\cdot P(k_i)}. \tag{16}$$

The knowledge fusion methods based on the Bayesian model can eliminate conflicts through probabilistic reasoning, but it relies on the independence assumption of knowledge sources and the accuracy of prior probabilities, and its application is limited in heterogeneous data source scenarios.

3.6.2. Fusion based on D-S Evidence Theory

D-S evidence theory [127] is a further extension of the Bayesian probability method, which describes the uncertainty of knowledge through the belief function (Bel) and the likelihood function (Pl). Specifically, for the proposition $A \in 2^{\Omega}$, where Ω represents the recognition framework composed of mutually incompatible basic propositions, the degree of trust in A can be obtained through the belief function $Bel(A)$. The likelihood function $Pl(A)$ represents the degree of trust in the non-false proposition A, that is, $Pl(A) = 1 - Bel(\neg A)$. Among them, the belief interval $[Bel(A), P_l(A)]$ describes the uncertainty range of proposition A. For the Basic Probability Assignment (BPA) m_1 and m_2 of two independent evidence sources A

and B, the combined BPA can be obtained by Dempster's combination rule, that is:

$$m(C) = \frac{\sum_{A_i \cap B_j = C} m_1(A_i) m_2(B_j)}{1 - k}, \tag{17}$$

where $C \neq \emptyset$, $k = \sum_{A_i \cap B_j = \emptyset} m_1(A_i) m_2(B_j)$ is the conflict coefficient used to measure inconsistency. The knowledge fusion method based on D-S evidence theory also requires that the knowledge sources involved in the evaluation are independent of each other, and when the knowledge sources are highly conflicting, contradictory conclusions are often produced. The time complexity of this method increases exponentially with the increase in the number of knowledge sources [127], and it is difficult to handle a large amount of knowledge fusion.

3.6.3. Fusion based on Graph Model

Graph model methods transform the problem of multi-source knowledge fusion into the optimization or reasoning problem of graph structure by modeling the association structure between knowledge, and have become the mainstream direction in recent years. Lao et al. [128] proposed a knowledge representation algorithm based on path sorting, which uses the random walk of the relationship path between entities to generate the probability of knowledge association. By enumerating the relationship paths in the knowledge base, the logistic regression model is trained to predict implicit knowledge. This method regards knowledge fusion as a link prediction problem on the graph and is suitable for relationship completion of structured knowledge bases.

Dong et al. [129] represent the knowledge base as a three-dimensional matrix ($G : E \times P \times E$), where E represents the number of entities and P is the number of predicates. The low-dimensional vector representation of entities and relations is learned through low-rank decomposition and a multi-layer perceptron:

$$Pr(G(s, p, o) = 1) = \sigma\left(\beta^T f\left[A\left[u_s, w_p, v_o\right]\right]\right), \tag{18}$$

where A is a $L \times 3K$-dimensional matrix used to represent the weights of the first layer of the multi-layer perceptron, and beta is a $L \times 1$ vector used to represent the weights of the second layer. By embedding entities and relations into the same semantic space, the data sparsity problem can be effectively handled.

In view of dynamic knowledge evolution, Zhao et al. [130] proposed the Gaussian Truth Model, which uses Gaussian distribution to model the entity truth μ_e and the quality of the data source s:

$$p\left(o, \mu, \sigma^2 \mid \mu_0, \sigma_0^2, \alpha, \beta\right), \tag{19}$$

where μ is the mean value, σ is the standard deviation, α is the shape parameter, β is the scale parameter, and o is the normalized value. The optimal estimate is solved iteratively through the EM algorithm:

$$\hat{\mu}_{MAP} = \arg max_{\mu} \int p(o, \mu, \sigma^2 | \mu_0, \sigma_0^2, \alpha, \beta). \tag{20}$$

This method explicitly models the time dimension of knowledge and the reliability of data sources, and is suitable for real-time fusion of dynamic multi-source knowledge.

In medical vision scenarios, multi-source knowledge fusion algorithms are widely used in knowledge graph data fusion and analysis, which requires balancing heterogeneous data compatibility, dynamic evolution modeling, and computational efficiency. Multi-source knowledge fusion algorithms combined with deep learning [131, 132] have also become a hot trend. How to combine deep learning algorithms with the more interpretable knowledge fusion methods introduced in this section is an important research direction.

4. Different Stages of Knowledge Fusion

In this section, we introduce the four knowledge fusion stages in the medical vision-language model, including data construction stage, pretraining stage, feature fusion stage and inference stage. The illustration of knowledge fusion in the four stages is shown in Figure 12.

4.1. Knowledge Fusion in Data Construction

For medical vision-language models, it is necessary to train the model based on a large amount of data. In the training data construction stage, knowledge fusion is mainly reflected in the filtering and annotation of data. In view of the particularity of medical data, the source and scope of the data will be determined based on medical expertise to ensure that the collected data covers representative and clinically significant medical images, case reports and other information. When collecting multimodal knowledge, we will focus on the typical characteristics corresponding to common diseases, and at the same time, combine anatomical knowledge to classify and organize data from different parts so that the model can learn knowledge related to the human body structure [14, 17].

In order to further enhance the model's ability to understand and utilize medical knowledge, some studies have begun to explore the deep fusion of multi-source heterogeneous medical knowledge. Based on the training dataset, they constructed a medical knowledge graph for specific tasks, and associated various medical concepts such as diseases, symptoms, examination methods, treatment plans and their interrelationships to provide a more comprehensive knowledge system support for the model [12, 133]. In addition, the large language model has demonstrated excellent natural language processing capabilities and is widely used in dataset processing. Existing methods are usually based on medical knowledge bases such as PubMed, UMLS, and WikiData, and use advanced large language models to construct training datasets or filter low-quality data [16, 134]. Some work [8, 135] are based on medical standards or medical papers, and reformulated them to obtain medical question-answering data with prior knowledge for fine-tuning or evaluation.

Compared with other stages, knowledge fusion in data construction stage can bring the following benefits. First, knowledge fusion can be more flexible, and the knowledge source can be dynamically adjusted according to specific task requirements without modifying the model architecture. Second, fusing knowledge at the data level can reduce training costs and

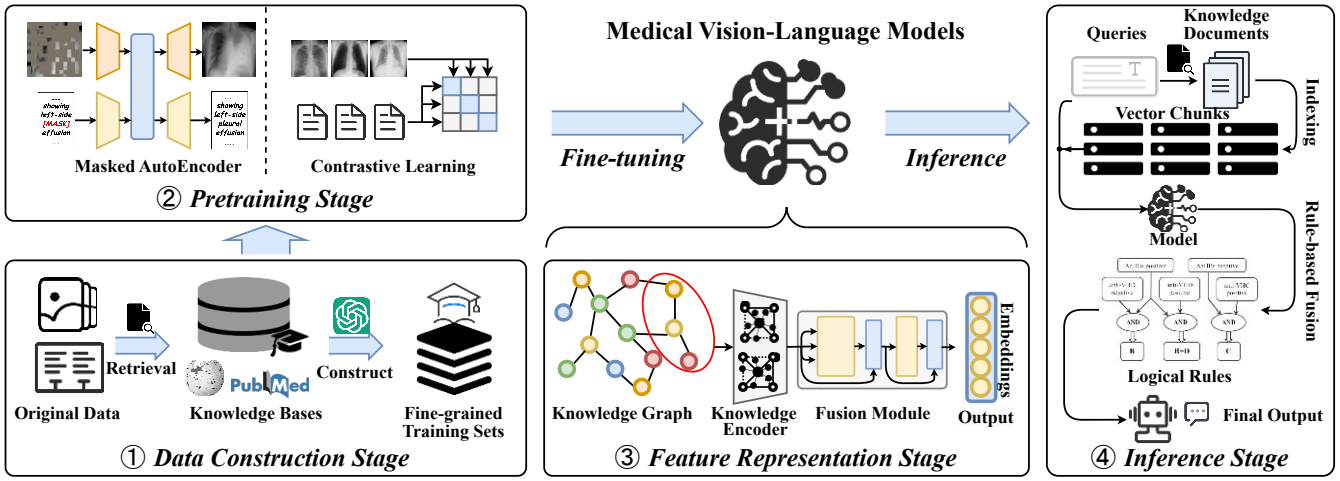


Figure 12: An illustration of four stages in knowledge fusion.

achieve better performance on smaller data. Finally, by carefully designing the knowledge source of data construction, the data distribution can be balanced to avoid the problem of long tail of medical data. The advantages and disadvantages of knowledge fusion in data constrcution stage are listed in Table 2.

4.2. Knowledge Fusion in Pretraining

Pretraining is the process of self-supervised training on large-scale unlabeled data before the model performs downstream tasks, so as to provide better initial parameters and feature extraction capabilities for downstream tasks. For medical vision-language models, in the pretraining stage, the prior knowledge in the medical field is introduced at the beginning of model training to enhance the model's understanding of medical concepts and relationships. The knowledge fusion in this stage is usually achieved through the design of pretraining tasks. Through these tasks, the model can learn rich medical knowledge from a large amount of medical text and image data, so as to show better performance in subsequent fine-tuning tasks. The pretraining stage is not limited to text data, but can also be combined with medical imaging data. Through multimodal pretraining tasks, the model can understand the medical information in text and images at the same time [4].

Unlike knowledge fusion in the data construction stage, pretraining focuses on large-scale unlabeled data to learn common feature representations and other knowledge. Many works introduce mask modeling as the goal of pretraining tasks to complete key entities in medical texts or restore masked areas of medical images [21, 100, 101]. Another important pretraining method is to improve the representation ability of features within or between modalities through contrastive learning by setting positive and negative samples [19, 20, 108, 111]. In addition, large-scale pretraining combined with more refined knowledge such as knowledge graphs is also an important research direction of medical vision-language models [10].

Compared with other stages, knowledge fusion in pretraining stage has the following benefits. First, it can introduce rich medical knowledge in the early stage of model training. Second, through large-scale pretraining, the model can learn more general medical representations that are suitable for a variety of downstream tasks. Finally, knowledge fusion in the pretraining stage can reduce the dependence on labeled data and reduce data labeling costs. The advantages and disadvantages of knowledge fusion in pretraining stage are listed in Table 2.

4.3. Knowledge Fusion in Feature Representation

Knowledge fusion in the feature level is the most important knowledge fusion strategy. It significantly enhances the model's ability to understand medical data by introducing prior medical knowledge during the feature extraction process. The core of this fusion method is to embed medical knowledge into the underlying structure of the model in the form of features, so that the model can capture key information more efficiently and accurately when processing complex medical data.

In practical applications, knowledge fusion at the feature level is usually achieved by designing a specific external knowledge feature extraction module. For example, in the CNN or Transformer architecture, a medical knowledge-guided attention mechanism can be added [22, 136]. This strategy can guide the model to pay more attention to medical-related areas when extracting features. The model can automatically focus on the lesion area or key anatomical structure, thereby improving the accuracy and efficiency of diagnosis. For knowledge graph-type data, feature extraction methods based on translation or graph neural network are currently mainly used. The translation-based strategy can embed entities and relationships in the knowledge graph into a low-dimensional vector space and further capture the structural relationship between them in this space [82, 137]. Graph neural networks can more naturally process graph structural relationships in knowledge graphs and capture long-distance dependencies between nodes [12, 81]. In addition, there are many works that directly learn specific knowledge representations from the teacher model through knowledge distillation, and embed the knowledge into the student model [121–123]. This method can embed complex medi-

cal knowledge into a lightweight student model, thereby reducing the computational cost while ensuring model performance.

Knowledge fusion in the feature level is currently the most commonly used fusion strategy, and its advantages are mainly reflected in the following aspects. First, it can directly introduce medical knowledge in the feature extraction process to make the feature representation of the model more medically meaningful. Second, through knowledge-guided feature extraction, the sensitivity of the model to noisy data can be reduced and the robustness of the model can be improved. Third, this stage can be combined with other stages to form a multi-level knowledge fusion system. Finally, knowledge fusion at the feature level is not only applicable to text data, but can also be applied to multiple modalities such as images and graphs, and is more suitable for multimodal knowledge fusion. The advantages and disadvantages of knowledge fusion in feature representation stage are listed in Table 2.

4.4. Knowledge Fusion in Inference

Knowledge fusion during the inference stage refers to the integration of additional medical knowledge into the model's prediction and decision-making processes after the model has been trained, thereby optimizing its inference process and output results. This integration allows the model to fully leverage its existing knowledge reserves when facing real-world medical problems, leading to more accurate and medically logical judgments [138].

The most common method for knowledge fusion during the inference stage is retrieval-augmented generation (RAG). The core of RAG lies in combining the retrieval and generation processes, enhancing the model's generation capabilities by retrieving relevant knowledge. In the medical field, some work [13, 24] use RAG to search for information related to the input data from vast repositories of medical literature, clinical case databases, and medical knowledge graphs, providing richer knowledge support for model inference. Additionally, the expertise of medical professionals can also be incorporated into the inference process. By transforming experts' diagnostic thinking and decision-making rules into computable forms, the model can draw on this experience during inference [139]. Another approach is rule-based reasoning, which encodes medical knowledge in the form of rules [26, 46, 124, 138]. During inference, the model matches these rules against the input data to derive corresponding conclusions.

Compared with other stages, knowledge fusion in the inference stage has unique advantages. First, this approach allows the model to quickly adapt to new knowledge and requirements without altering its structure or retraining. Second, knowledge fusion during inference enhances the model's interpretability. Since the model explicitly references external knowledge during reasoning, its decision-making basis becomes clearer and more understandable to both medical professionals and patients. Finally, it helps improve the model's generalization ability. When confronted with complex and variable clinical data, the model can better handle various situations by integrating knowledge and enhancing its performance across different scenarios. The advantages and disadvantages of knowledge fusion in inference stage are listed in Table 2.

5. Knowledge Fusion in Medical Vision-Language Tasks

This section mainly introduces knowledge fusion in five common medical vision-language tasks, including medical report generation, medical visual question answering, medical language-guided segmentation, medical multimodal pretraining, and medical multimodal large language model. The overview of this section is shown in Figure 13.

5.1. Medical Report Generation

Medical reports are an integral part of the healthcare process and are usually written manually by experts based on medical images. Report writing is time-consuming and is influenced by the doctor's prior knowledge. Medical report generation is automated through multimodal technology. However, medical reports involve complex professional terminology and logical reasoning processes, and purely data-driven models may generate erroneous or inconsistent content. Therefore, medical report generation based on knowledge fusion has become a hot topic.

The additional disease labels contained in the report are important external knowledge in the report generation task. Multi-task training based on these labels is a common knowledge fusion method. In practical applications, disease labels can be used to guide the generation process to focus on specific semantic areas, thereby generating more accurate and medically meaningful reports [8, 140–143]. Li et al. [140] were the first to propose combining medical knowledge for medical report generation and introduced a knowledge retrieval process into the model. They manually designed a disease graph and an abnormality graph, integrating the visual features obtained from the image encoder into the graph structure. Text templates were then retrieved through the retrieval module based on the detected abnormality and finally rewritten for specific cases by the rewriting module. Liu et al. [142] simulated the process of radiologists writing reports by extracting posterior and prior knowledge. The model consists of three similar Transformer modules: PoKE, PrKE and MKD, where PoKE is responsible for associating disease topic labels with abnormal areas, and PrKE is responsible for fusing the text features of clinical reports with prior image features. Jia et al. [143] constructed a static disease-correlation matrix and used the co-occurrence relationship between disease labels to achieve knowledge transfer from common diseases to rare diseases. They further achieved multi-modal alignment through a multi-head attention mechanism and generated reports by fusing key features of different modalities.

Knowledge graph is a major source of knowledge in medical report generation. By organizing the concepts, relationships, and attributes in the medical field into knowledge graphs, the model can better understand the semantic relationships in medical reports [12, 81, 82, 84, 133]. Li et al. [133] used tools to automatically construct knowledge graphs and decoupled the report generation task into two stages: triple reconstruction and natural language generation. Similarly, Xu et

Table 2: The advantages and disadvantages of knowledge fusion in four stages.

Stages	Advantages	Disadvantages	Papers
Data Construction	Dynamically adjust Reduce training costs Balance data distribution Avoid long tail problems	Oversimplification Missing context High labeling costs Low timeliness of data	[8, 12, 14, 16, 17, 133–135]
Pretraining	Introduce rich medical knowledge Learn more general medical representation Reduce data labeling costs	Misalignment More noise Solidification	[4, 10, 19–21, 100, 101, 108, 111]
Feature Representation	Directly introduce knowledge Enhance robustness Integration with other stages Multimodal knowledge fusion	Solidification Information loss Overfit Limited generalization	[12, 22, 81, 82, 121–123, 136, 137]
Inference	Enhance generalization ability Quickly adapt	High complexity High dependence on logic	[13, 24, 26, 46, 124, 138, 138, 139]

al. [12] used graph neural networks to encode prior knowledge graphs, classified prior knowledge labels, and then retrieved and reconstructed subgraphs in the knowledge graph to obtain fine-grained knowledge to assist in the generation of skin disease reports. Yang et al. [82] proposed a medical report generation architecture that combines general knowledge and specific knowledge. They used RadGraph [144] as the knowledge graph of radiology images, obtained general knowledge through global knowledge graph representation learning, and obtained similar reports as specific knowledge through image similarity retrieval, which were fused and input into the decoder. Huang et al. [84] combined clinical knowledge with visual and contextual information by using symptom maps constructed by professional doctors, and performed knowledge distillation in the decoding stage. They designed a Transformer with a U-connected architecture to enable interaction between different modalities and knowledge.

As the model's text understanding ability improves, unstructured text expert knowledge bases have become a better choice. Compared with structured knowledge graphs, unstructured text can more flexibly express complex semantics and diverse medical knowledge [145–148]. Liu et al. [145] built an M-VLBert model and designed a term encoder based on a pretrained model that can fuse features of images, medical terms, and medical reports. Medical reports of CT images are generated by a shared Transformer decoder. The decoder adds an auxiliary language module, extracts contextual language features using the Bert model, and fuses visual features and contextual language features through multi-head attention in the final report generation. Yang et al. [146] built a learnable knowledge base that can automatically extract and recover medical knowledge from text embeddings and fuse its visual features and text features through visual-knowledge attention. Bu et al. [147] proposed a framework called EKAGen, which converts expert reports into an embedding space and generates a comprehensive representation for each disease as preliminary knowledge support. EKAGen creates instance-level expert knowledge for each query image by diagnosing the disease and retrieving knowledge from the correct knowledge support to facilitate report generation. Bu et al. [149] proposed a dynamic instance-level explorer to dynamically assign instance-level prior knowledge prompts to each image from the collected report knowledge base to strengthen the description of abnormal areas; and proposed a prior knowledge prompt fuser to fuse dynamic knowledge prompts with hidden states to create enhanced multimodal features, thereby enhancing generation capabilities.

These knowledge fusion strategies have significantly improved the quality and efficiency of report generation in the field of medical report generation. From the perspective of quantitative evaluation metrics, Li et al. [140] achieved a 10%-15% improvement in automatic evaluation metrics compared with traditional data-driven models by decoupling the report generation task based on knowledge graphs, verifying the advantages of knowledge in semantic consistency and medical accuracy. PPKED [142] improved the ROUGE-L score by 9.9% compared with the baseline model when processing complex imaging reports, especially in terms of the accuracy and logical coherence of key medical terms, proving the effectiveness of knowledge fusion strategies in complex semantic expressions.

5.2. Medical Visual Question Answering

Medical visual question answering is an important research area in the field of intelligent medicine. It combines professional knowledge and visual language models to improve the model's understanding of medical images and the accuracy of question answering. Traditional medical visual question answering models [150–152] mostly adopt a pure data-driven approach, using deep learning technology to extract features from images and texts and generate answers. However, there are complex semantic structures and logical reasoning requirements in medical scenarios. Data-driven methods are easily limited by the scale and quality of training data, which may lead to a lack of professionalism and consistency in the generated results. Therefore, injecting external knowledge into the model and building a medical visual question answering model based on knowledge enhancement has become a mainstream research direction.

In the early days, due to limitations on model parameters and data set size, some models considered building knowledge templates or converting questions into more fine-grained classifications to facilitate knowledge retrieval and generate relevant responses, thereby simplifying the difficulty of task learning [153, 154]. Kornuta et al. [153] defined the knowledge in medical visual question answering as the information learned

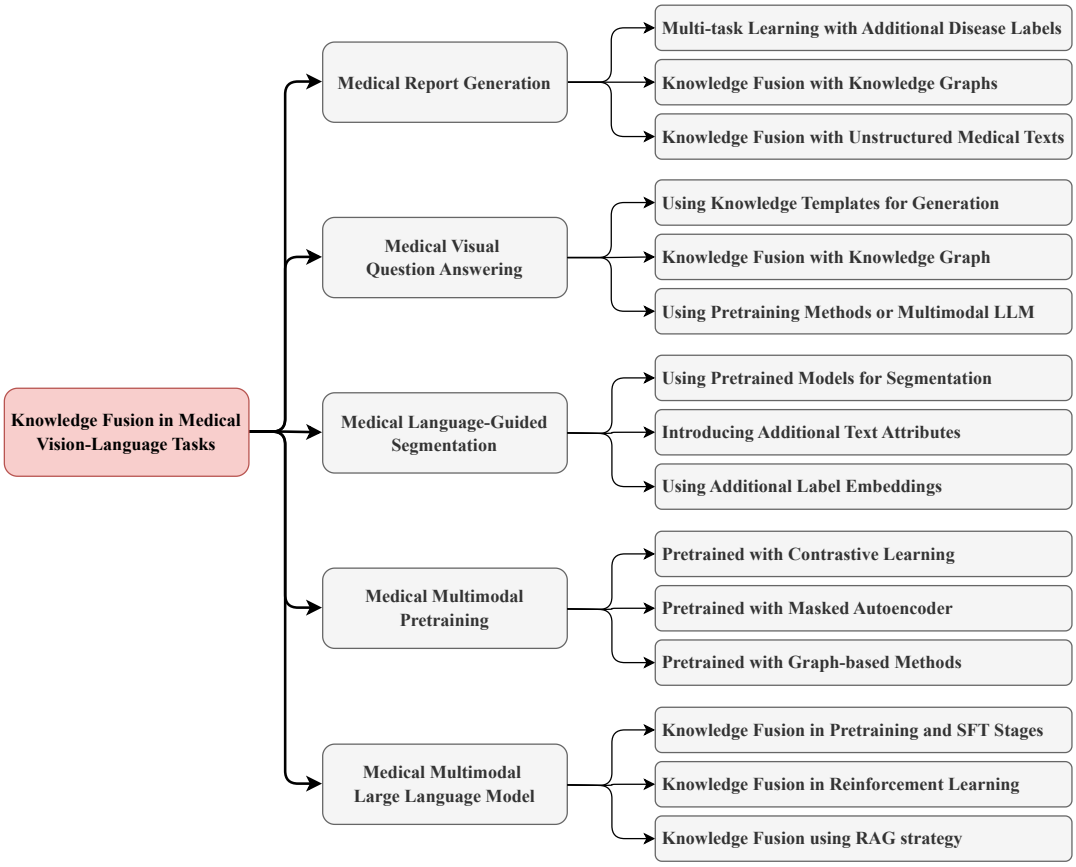


Figure 13: The overview of knowledge fusion in medcial vision-language tasks.

from the upstream task, using them as supporting facts to help solve more complex downstream tasks. The model includes a supporting network, and each subtask includes a corresponding classifier to decouple the answers in different situations. Liao et al. [154] used the Skeleton-based Sentence Mapping (SSM) method to map questions to a set of skeleton structures, thereby inferring the question category and corresponding labels, and transforming the visual question answering task into a multi-task image classification problem. In the question understanding and image modeling stages, the SSM method is used to transform the question into a classifiable task, and then feature fusion and knowledge reasoning are performed in the image classification model.

Similar to report generation, knowledge graphs are also used in medical visual question answering [9, 155–157]. Zheng et al. [155] constructed a knowledge graph to represent the relationship between images and text, used a graph neural network to propagate node information, and generated feature representations of the knowledge graph. Liu et al. [156] built a dataset with rich semantic annotations and medical knowledge graphs to support medical visual question answering. The knowledge graph is centered on organs and related diseases and contains a large number of medical knowledge triplets to help the model understand complex medical questions. Hu et al. [157] combined semantic knowledge and spatial knowledge to build an expert knowledge graph, and focused on the prior knowledge of anatomical structures to capture the detailed differences between images for visual question answering. Jian et al. [9] built a medical knowledge graph that includes entities such as diseases, symptoms, examination items, and treatment methods based on the types of diseases and questions that patients are concerned about during treatment. During training, images, questions, and knowledge graphs are fused through three different fusion modules, and the joint embedding of different modules is interactively learned to assist in subsequent medical visual question answering.

As the number of available corpora increases, more and more models are improving the performance of medical visual question answering through pretraining [18, 123, 158–160]. Wang et al. [123] use a multimodal hierarchical knowledge distillation method to learn rich representations of medical images and texts, and transfer the knowledge of complex teacher models to simpler student models, thereby reducing the complexity of the model while maintaining high performance. Li et al. [158] combined multiple self-supervised strategies such as mask modeling tasks, image-text matching tasks, unimodal contrastive learning, and multimodal contrastive learning for training. After pretraining, the learned feature representations were transferred to the downstream medical visual question answering task by adding a decoder to generate answers. He et al. [159] proposed a robust pretraining strategy for medical visual question answering combined with contrastive learning. By adding adversarial perturbations to the image, the model can better cope with complex noise in the data during the learning

process and improve the generalization ability of the model.

The emergence of large language models breaks the limitation of medical visual question answering in the past, which was to select answers through classifiers, and truly achieves visual question answering through generative methods. Zhang et al. [161] constructed a PMC-VQA dataset containing 227k question-answer pairs based on the PubMed knowledge base, and used this knowledge data to train the medical visual question answering model MedVInT. The goal is to generate open-ended natural language answers instead of selecting from a predefined answer library. Yan et al. [162] used a large-scale medical image-text dataset and the Unified Medical Language System to build a knowledge graph containing 15,635 medical entity nodes and 130,196 relationship edges. The knowledge graph was encoded using a graph attention network to capture external medical knowledge.

Knowledge fusion approaches for medical visual question answering have been shown to have a significant impact on model performance. Models that integrate knowledge graphs proposed by Liu et al. [156] and Hu et al. [157] have achieved significant improvements in accuracy compared to traditional data-driven models. By leveraging the structured medical knowledge embedded in the graphs, these models are able to better understand the context of medical questions, leading to more accurate and relevant answers.

In addition, knowledge fusion strategies have significance in clinical practice. These models can help doctors quickly obtain detailed information from medical images. Using knowledge-infused pretrained models [123, 158] can explore new medical knowledge by analyzing large-scale medical image text data. These models can discover hidden patterns and relationships in medical data, helping to develop new diagnostic methods and treatment strategies.

5.3. Medical Language-Guided Segmentation

Traditional medical image segmentation tasks typically take medical images as input and use deep learning models to directly generate detection results from the images. However, with the development of multimodal research, it has become increasingly evident that information from a single modality often fails to adequately express the complexity of medical knowledge. As a result, studies have emerged that incorporate textual input for visual segmentation tasks, i.e. language-guided segmentation. Based on the varying roles of textual input in the task, such research can be further categorized into the following types: (1) Medical Image Segmentation: Text serves solely as additional information to enhance segmentation results, so this task remains within the scope of traditional lesion segmentation. (2) Medical Image Referring Segmentation. The medical image is accompanied by textual descriptions that identify specific lesions within the image. Referring focuses on providing textual descriptions of specified regions in the image and outputing text-related segmentations. (3) Referring and Grounding. With the development of pretrained large language models (LLMs), research have begun leveraging the powerful text processing capabilities of LLMs, enabling models to simultaneously perform both referring and grounding tasks. Grounding involves segmenting specific regions and linking language with visual elements. Once a model possesses both referring and grounding capabilities, it can accurately reference specific regions for detailed diagnosis and report back to the correct areas, thereby enhancing interpretability. Recent studies [23, 163, 164] have combined textual and visual knowledge to enhance the model's understanding of complex medical knowledge and effectively improve the robustness of segmentation tasks.

Pretrained models are trained on large-scale datasets and contain a wealth of knowledge. Mining the existing knowledge within pretrained models can effectively leverage this prior knowledge, thereby reducing the need for model training. Feng et al. [165] established a connection between the textual diagnostic information and the intermediate activations in the reverse diffusion process of the Diffusion Probabilistic Model (DPM). Specifically, they designed a cross-modal attention-based knowledge alignment within the DPM. The cross-modal attention module provided a strong link between language and vision, enabling the textual information to enhance the semantic representation within the image. Li et al. [166] utilized external knowledge contained in textual descriptions by preprocessing it through a robust image-text knowledge embedded in vision-language models such as CLIP [104], ultimately generating an explicit prior. The pretrained CLIP was used as the text encoder, while the pretrained segmentation anything model (SAM) [167] served as the image encoder. Both encoders were frozen, and the obtained external knowledge was injected into each layer of the SAM image encoder using a prior-aligned injector. This approach facilitated knowledge sharing between the segmentation and vision-language models. Inside the prior-aligned injector, cross-modality interaction was implemented using an improved attention mechanism, combined with residual connections to enhance stability.

Textual attributes, such as descriptive key properties included in medical concepts (e.g., color, shape, size), contain expert-level textual knowledge that can significantly enhance the understanding and interpretation of visual data [28, 168–170]. En et al. [168] introduced a textual attribute knowledge learning module that extracts attribute knowledge and integrates it into feature representations through auxiliary attribute classifications. Text attributes were obtained from text descriptions, while image embeddings were derived from the input image, leading to the generation of a prediction mask. The predicted mask was then used to obtain classification results, which were compared with the text attributes to compute a classification loss. Qin et al. [28] designed an auto-prompt generation method to automate the creation of prompts. These prompts include expert-level textual knowledge, such as descriptive key attributes within medical concepts. These attributes are shared between general and medical domains. Therefore, the method leverages the natural image knowledge learned from pretrained general vision-language models and the expert-level medical textual knowledge embedded in pretrained medical language models. To enable the model to learn various features of the required anatomical structures, Kim et al. [169] retrieved text de-

scriptions of different labels from internet resources. A knowledge encoder was used to convert the text descriptions into knowledge features, which guide the segmentation model to accurately identify the required labels. The knowledge feature was then average-pooled along the sequence axis and processed through an adaptation layer, producing a modified knowledge feature.

Current work typically uses one-hot or few-shot labels for image categories, which fail to carry semantic knowledge and anatomical relationships between categories. However, these relationships between medical entities represent one of the key sources of medical knowledge. Liu et al. [171] used the pretrained text encoder of CLIP, to convert labels into embeddings. The converted CLIP embeddings contained relationships between medical entities, such as the anatomical relationships between organs. Similarly, Liu et al. [172] proposed a comparable label modeling approach. They introduced a language-driven parameter generator module, which employed an independent multilayer perception (MLP) to output a parameter θ for each category based on the language embedding. The class-specific head then predicted the binary mask using the learned θ. To add new categories, the generator module could be extended by adding new MLPs. In the expanded dataset, pseudo-labels were used for old data, and ground truth for new data, enabling the new knowledge to build upon the old.

The advancements in knowledge fusion for medical language-guided segmentation, as discussed, offer significant potential to practical medical applications. For instance, in auxiliary diagnosis, the ability to guide segmentation with precise textual commands, which leverages fused knowledge from textual attributes can lead to more accurate and consistent delineation of pathologies. The integration of knowledge fusion has demonstrably elevated the capabilities of medical language-guided segmentation. By incorporating diverse knowledge sources, sourced from textual attributes defining lesion characteristics to semantic embeddings of anatomical relationships, models now achieve more precise and contextual-aware delineations. This has also caused shifts in training paradigms, such as leveraging large pre-trained vision-language models (e.g., CLIP [104], SAM [167]) as powerful knowledge priors, enabling more effective few-shot or even zero-shot segmentation and reducing the reliance on extensive, task-specific annotated datasets.

5.4. Medical Multimodal Pretraining

With the explosion of medical image and text data, multimodal pretraining models play an increasingly important role in knowledge fusion-based medical vision-language tasks. They provide powerful support by realizing deep image and text fusion through different technical approaches and architectural designs. The pretraining stage of medical vision-language models is the cornerstone of building a high-performance system, and its core goal is to achieve deep semantic alignment and knowledge fusion of medical images and texts through the learning of large-scale multimodal data. In recent years, researchers have proposed a variety of innovative multimodal pretraining frameworks to address the challenges of strong data heterogeneity, common modality missing, and complex knowledge fusion in the medical domain, which have significantly advanced the development of medical multimodal representation learning.

For medical multimodal pretraining models, a more natural idea in dealing with cross-modal tasks is based on the idea of cross-modal alignment and contrastive learning [19, 20, 108, 111, 173, 174]. Huang et al. [19] proposed a multimodal global-local representation learning framework, GLoRIA, which aims to improve the efficiency of labelling for medical image recognition by using radiological reports. The framework proposes a bi-level comparison objective that simultaneously models the overall semantic alignment of image-text and the local semantic association of region-keywords to enhance the model's sensitivity to medical image details. Sharma et al. [173] proposed a lightweight multimodal pretraining framework, MedFuseNet, which aims to solve the problems of low efficiency of multimodal alignment and high model complexity in medical visual question answering. Specifically, the framework reduces the redundant computation of the traditional two-stream model by adopting a parameter-sharing cross-modal attention mechanism, and introduces a dynamic gating module to adaptively fuse visual and textual features to improve the modeling efficiency. Liu et al. [174] developed a new vision-language pretraining framework called Imitate. It extracts multi-level visual features from chest X-ray images and aligns these features with the descriptive content and concluding text in the medical report, respectively, through a hierarchical alignment mechanism, which enables the framework to better understand the structured information of the medical report. Wu et al. [20] proposed the MedKLIP framework for integrating medical knowledge into image and text alignment to enhance X-ray diagnostic accuracy. It contains a triple extraction module and combines knowledge-enhancing ternary coding.

Another common practice in medical multimodal pretraining models is based on mask reconstruction with mixed targets [21, 100, 101, 175]. Chen et al. [100] proposed M3AE and solved the problem of sparse medical multimodal data and complex domain knowledge through joint masked reconstruction and cross-modal alignment. Liu et al. [21] used M3AE to solve the missing modalities problem encountered in brain tumor segmentation tasks. They also proposed a memory-efficient self-distillation method for knowledge transfer in different missing modality situations. It reduces the number of parameters required by using a single network model for all possible combinations of modal deficiencies. Moon et al. [101] proposed a model called medical vision-language learner model. It uses a BERT-based architecture and incorporates a novel multimodal attention masking scheme. This architecture aims to maximize the generalization performance of vision-language tasks, such as diagnostic classification, medical image-report retrieval, and medical visual question answering.

Graph-based methods also play an important role in the recent rise of pretraining methods [10, 176, 177]. Peng et al. [176] proposed a novel cue learning model for the diagnosis of neurological disorders by learning graphical cues during the fine-tuning of multimodal models. Specifically, it first utilizes GPT-

4 [178] to obtain relevant disease concepts and computes the semantic similarity between these concepts and all image blocks. In terms of other innovative points, Sengupta et al. [177] proposed the model which is not only brought closer to the target domain, but is also familiarized with the specific knowledge of that domain by means of goal-oriented pretraining. Chen et al. [10] proposed a systematic approach to enhance medical vision-language pretraining through structured medical knowledge.

The introduction of knowledge fusion strategy has significantly revolutionized the technical paradigm of medical multimodal pretraining. Overall, the contrastive knowledge fusion framework significantly improves cross-modal retrieval metrics, with an average precision of 6.8% [179] on benchmarks such as MIMIC-CXR [38] compared to the baseline. Graph-based cue learning and knowledge-enhanced pretraining methods can accelerate convergence, reduce pretraining time by about 15-20%, and improve VQA and report generation accuracy by more than 3% [180–183]. Finally, the hybrid training paradigm combined with multiple tasks standardizes the downstream fine-tuning workflow and promotes the deployment of multimodal models in clinical settings. Future research needs to focus on building a scalable meta-learning framework and establishing a dynamic knowledge distillation mechanism to further improve the effect of knowledge fusion during pretraining.

5.5. Medical Multimodal Large Language Models

Since the release of ChatGPT, large language models have achieved unprecedented success in the field of natural language processing. The subsequent release of GPT-4 [178] marked a significant milestone, as it featured excellent multimodal capabilities, including visual recognition. This model can easily identify the content of images uploaded by users and provide reasonable answers. GPT-4V further demonstrated its potential by achieving an overall accuracy of 90.7% on the US Medical Licensing Examination, surpassing the performance of most medical students, especially on image-based questions. This remarkable achievement can be attributed to the model's integration of vast amounts of high-quality multimodal knowledge within its internal structure. Following this success, more medical multimodal large language models have been released, with continuous innovations in design.

For multimodal large language models, the first step is preparing a dataset for pretraining, as a wealth of implicit medical multimodal knowledge is embedded in the raw corpus [14–17, 135, 184]. Li et al. [14] proposed the LLaVA-Med model, which was trained on a highly diverse visual instruction dataset created by sampling biomedical image-text pairs from PMC-15M and using GPT-4 to generate instructions from the text, thereby creating a variety of instances. This dataset covers all findings in biomedical imaging and provides a diverse knowledge source for model training. Huang et al. [16] designed the Med-GRIT-270k dataset, which covers eight different medical imaging modalities. Through this dataset and multitask instruction learning, they introduced biomedical references and the general multimodal large language model, validating the effectiveness of multitask instruction fine-tuning and achieving the fusion of domain-specific knowledge with the model. Chen et al. [135] improved PubMed's medical image-text pairs, innovatively using "non-blind" capabilities of GPT-4V to denoise and reformat data, creating the PubMedVision dataset with 1.3 million medical VQA samples, providing rich medical visual knowledge for subsequent training. Liu et al. [184] proposed the Qilin-med-vl model and released ChiMed-VL, a carefully curated dataset of over 1 million image-text pairs. This dataset, leveraging various types of images, provides detailed and comprehensive explanations of medical data, thereby supporting model training in the Chinese medical field. In practice, full pretraining is often not required, and a more common approach is continuous pretraining, which involves further training the model with new relevant data based on an already pretrained model, allowing it to adapt to new knowledge and task requirements. In the medical domain, Med-Flamingo [185] also makes important contributions to dataset preparation. It constructs a unique, interleaved image-text dataset from over 4K medical textbooks, along with using the PMC-OA dataset. The dataset, consisting of approximately 0.8M images and 584M tokens, is carefully curated from authoritative medical knowledge sources. This dataset enables the pretraining of a multimodal few-shot learner for the general medical domain, unlocking few-shot generative medical visual question answering abilities.

After the pretraining stage, supervised fine-tuning is used to adjust the pretrained model based on specific task and domain data, optimizing its performance on knowledge integration for the target task and domain [134, 186–188]. Thawakar et al. [186] proposed the XrayGPT model, which combines a medical visual encoder with the fine-tuned LLM, enabling visual dialogue based on the understanding of X-ray images and medical knowledge. This fusion of medical visual and textual knowledge through fine-tuning enabled the completion of chest X-ray summarization tasks. Zhu et al. [188] proposed the universal medical model Uni-Med, which consists of a general visual feature extraction module, an expert connector mixture module. Supervised fine-tuning enables it to perform six different medical tasks, including question answering, visual question answering, report generation, facial expression understanding, expression generation, and image classification, achieving the fusion of knowledge across various medical tasks. Xu et al. [134] introduced the innovative multi-level vision-language model Mlevlm, which represents a significant advancement in the field. They designed a multi-level feature alignment module to align features from visual to semantic levels, improving the model's performance in medical visual question answering tasks and facilitating the fusion and application of multi-level knowledge.

After the supervised fine-tuning stage, reinforcement learning with human feedback aims to optimize the model's behavior by introducing human feedback, aligning the knowledge embedded in the model's generated results with human expectations and preferences. This method further enhances knowledge integration in practical applications [189–192]. Hein et al. [190] proposed a scalable automated preference alignment technique that uses publicly available datasets and the LLM-

as-a-Judge mechanism, eliminating the need for additional radiologist feedback. This method optimizes model preferences in chest X-ray report generation tasks, improving the report's factuality and offering an innovative alternative to human feedback in traditional RLHF. Zhu et al. [192] proposed a new multimodal medical preference optimization method that considers the clinical relevance of preference samples, integrating clinical relevance as weights in preference optimization to achieve effective alignment and enhance model performance in clinical medical applications. In the research of reinforcement learning for large language models, Logic-RL [193] provides a new perspective. It explores the potential of rule-based reinforcement learning in large reasoning models. Using synthetic logic puzzles as training data, it designs a system prompt emphasizing the thinking and answering process, a stringent format reward function to avoid the model taking shortcuts, and a straightforward training recipe to achieve stable convergence. This framework helps the model develop advanced reasoning skills such as reflection, verification, and summarization, and shows good generalization ability on challenging math benchmarks.

Knowledge fusion can also be achieved through lightweight methods that do not require pretraining or fine-tuning of the model, such as Retrieval Augmented Generation (RAG) [13, 24, 25]. Xia et al. [13] proposed a two-part framework, RULE, which consisting of calibrating the selection of retrieved contexts to control factual risk and preference dataset fine-tuning to address errors caused by over-reliance on retrieved contexts. Xia et al. [24] proposed a general-purpose multimodal RAG system, MMed-RAG, which introduces a domain-aware retrieval mechanism, an adaptive retrieval context selection method, and a provable RAG-based preference fine-tuning strategy. Qu et al. [25] introduced the active retrieval-augmented large vision-language model to address hallucination issues from three key dimensions: accurately analyzing retrieval targets based on the inherent structure of images, exploring the most effective retrieval methods to select reliable results, and timing the retrieval process to match low-certainty events and avoiding unnecessary retrieval in high-certainty phases.

The incorporation of knowledge fusion strategies has significantly reshaped the training paradigms of medical multimodal large language models, enhancing both their adaptability and clinical performance. By leveraging domain-specific datasets, structured medical knowledge, and adaptive fine-tuning techniques, these models have demonstrated substantial improvements in downstream tasks. For example, the multi-level vision-language alignment module of Xu et al. [134] led to a 9.2% improvement in VQA accuracy by enriching semantic understanding across modalities. Similarly, preference optimization methods integrating clinical relevance, as proposed by Zhu et al. [192], improved factual correctness scores in radiology report generation by over 11%. These advances highlight how knowledge-guided fine-tuning and reinforcement methods not only reduce hallucinations but also align model outputs more closely with clinical expectations, paving the way for safer and more reliable deployment in medical settings.

5.6. Datasets

Advances in basic medical vision-language models heavily depend on the availability of diverse multimodal datasets. In the era of generalized artificial intelligence, large-scale datasets are crucial for developing robust medical-based models. The quality, size, and diversity of the data directly affect the generalization ability of these models, making pretraining of comprehensive datasets particularly important [194]. In the medical field, the accumulation of large-scale data provides a rich source of knowledge for training models that can perform different tasks in multiple domains.

In light of these challenges, it is imperative to develop ways to effectively utilize available multimodal data to improve the performance of medical vision-language models to better generalize between tasks and datasets. By maximizing the use of multimodal data, researchers can break through the constraints of the underlying models. In the following, we present the datasets categorized by task for medical knowledge fusion. Specifically there are four categories: Medical Report Generation (MRG), Medical Visual Question Answering (MVQA), Medical Language-Guided Segmentation (Lang-SEG) and Multimodal Pretraining (Pretrain). Datasets are shown in Table 3.

1) MRG: MIMIC-CXR [38] serves as the most common dataset in the field for training and evaluating the model's ability to learn multimodal representations. IU-Xray [195] is widely used to generate diagnostic reports based on images and to assess the clinical relevance of the generated text. PadChest [196] is originally reported in Spanish and needs to handle cross-language model training.

2) MVQA: Slake [156] contains 50 percent medical images in English and 50 percent in Chinese. The average length of RadQA [197] is more than 500 words, which aims to handle long distance dependencies problems. VQA-Med-2019 [198] involves radiology terminology and complex clinical reasoning. VQA-RAD [199] has a limited number of image-question-answer pairs and needs to cope with small sample learning problems. PubMedVision [200] is a large-scale, high-quality medical VQA dataset reformatted using GPT-4V. PathVQA [201] aims to handle large data sizes and generate answers that incorporate pathology terminology and diagnostic criteria.

3) Lang-SEG: MedTrinity-25M [202] adapts to both generative and discriminative tasks. MosMedData [203, 204] is designed to support COVID-19 diagnostic, lesion segmentation and disease analysis studies based on CT images. RefHL-Seg [164] aims to advance research on language-guided 3D medical image segmentation. QaTa-COV19 [205] is designed to support deep learning studies of new crown pneumonia images.

4) Pretrain: ROCO [206] could be used to build generative models for image captioning, classification models for image categorization and tagging or content-based image retrieval systems. The textual descriptions of MedICaT [207] are precise to lesion size, morphology, location. PMCVQA [161] has far more data than traditional medical VQA datasets, but is more

noisy. PMC-OA [208] is a major biomedical literature resource designed to support open science and text mining.

6. Evaluation of Knowledge Fusion

How to evaluate the medical vision-language model after knowledge fusion is crucial for ensuring the accuracy, reliability, and effectiveness of the model. In this section, we categorize the evaluation methods into three major types: quantitative evaluation, qualitative evaluation, and benchmark-based evaluation.

6.1. Quantitative Evaluation

Quantitative evaluation methods mainly measure the performance of the model after knowledge fusion through a series of metrics. These metrics include common metrics such as Accuracy, Recall, Precision, and F1 score. These metrics can comprehensively reflect the performance of the model in classification tasks.

For generative models, metrics such as the ROUGE [209] series and the BLEU [210] series are commonly used evaluation methods. ROUGE series metrics, such as ROUGE-N, ROUGE-L, ROUGE-W, evaluate the quality of text generation by calculating the n-gram overlap between the generated text and the reference text. BLEU series, such as BLEU-1, BLEU-2, BLEU-3, BLEU-4, metrics evaluate the quality of the model output by calculating the n-gram exact match rate between the generated text and the reference text. There are also two other metrics, METEOR [211] and CIDEr [212]. METEOR is based on the harmonic mean of precision and recall, with recall having a higher weight than precision. This metric is designed to correct some of the deficiencies of BLEU and shows good consistency with human judgment at the sentence or discourse level. CIDEr is a consensus-based evaluation criterion. Its basic principle is to evaluate the similarity of the sentence to be evaluated with human descriptions by measuring the similarity between the sentence to be evaluated and most other manually described sentences. Compared with the previous metrics, it has a higher consistency with human consensus. In addition, metrics such as Bert-sim [213], are also widely used. Bert-sim calculates the similarity between texts using the embedding vectors of the BERT model.

For vision-language models, especially in image segmentation tasks, DICE [214], mean Intersection over Union (mIoU) [215], and Average Precision at 50% Recall (AP50) are commonly used evaluation metrics. DICE is a metric used to measure the similarity between two sets. In medical image segmentation, the higher the DICE, the more similar the segmentation results generated by the model are to the true segmentation results. mIoU is one of the commonly used metrics for evaluating the performance of models in image segmentation tasks. It measures the segmentation accuracy of the model for different types of objects by calculating the Intersection over Union for multiple categories and taking the average. The AP50 evaluates the detection performance of the model by calculating the average precision of the model for all objects when the recall rate is 50%.

In addition to the above common metrics, Delbrouck et al. [216] proposed a new knowledge evaluation indicator, namely a reward mechanism based on the RadGraph dataset. It evaluates the factual accuracy and integrity of the generated report by calculating the differences in entities and relationships between the generated report and the reference report. Specific metrics include RG_E, RG_{ER}, and $RG_{\bar{ER}}$. These metrics can effectively capture the consistency of entities and relationships in medical reports, providing a new perspective for evaluating the knowledge fusion ability of medical vision-language models in radiology report generation.

6.2. Qualitative Evaluation

Qualitative evaluation methods mainly involve scoring and ranking the generated results after knowledge fusion through manual assessment or large language models, or conducting evaluations through methods such as error analysis and case studies.

Manual evaluation is a direct and subjective assessment method. Professional doctors or domain experts score and rank the texts generated by the model to evaluate the model's ability to understand and express medical knowledge. The evaluation factors include the relevance, accuracy, completeness, logical coherence, and interpretability of the text, etc [217]. Manual evaluation can fully consider the professional knowledge and practical application requirements in the medical field, but the evaluation results may be affected by the subjective factors of the evaluators. Using other large language models to evaluate the texts generated by medical vision-language models [134] is also an effective qualitative evaluation method. LLMs can provide evaluation results quickly and efficiently and can replace manual evaluation to a certain extent. However, attention should be paid to the differences and limitations among models.

Error analysis is an important part of qualitative evaluation. By analyzing the errors that occur during the model's generation process, the deficiencies of the model can be identified, and a basis can be provided for the improvement of the model [218]. Through the classification and analysis of error types, the problems and deficiencies of the model in the knowledge fusion process can be deeply understood. Data quality problems may lead the model to learn incorrect knowledge or information; insufficient model training may result in poor generalization ability and accuracy of the model; inadequate knowledge representation may prevent the model from accurately understanding and expressing medical knowledge.

Case study is another commonly used qualitative evaluation method. The knowledge fusion ability of medical vision-language models is evaluated through the presentation of specific cases [218]. The cases can cover different medical fields and tasks, such as disease diagnosis, treatment recommendations, medical image description, etc. Evaluate and analyze the effectiveness of the cases, including whether the texts generated by the model accurately and completely answer the questions, whether they conform to medical common sense and clinical practice, and whether they can provide effective reference and support for doctors.

Table 3: The summary of several datasets for different medical vision-language tasks.

Dataset Name	Modality	Tasks	Size	Disease/Anatomy	Description/Openness
MIMIC-CXR [38]	X-ray	MRG	377K images, 227K reports	Pneumonia, Pneumothorax, Pleural effusion, etc.	Published by Beth Israel Deaconess Medical Center. Openness:Apply for Access. License:PhysioNet Credentialed Health Data License 1.5.0.
IU X-ray [195]	X-ray	MRG	7.4k images, 3.9k reports	Nodule/Mass, Consolidation, Edema, etc.	Published by Indiana University. Openness:Open Access. License:MIT.
Open-i [10]	X-ray, CT, MRI, US, Pathology	MRG	8.1k images, 4k reports	Tumors, Cardiovascular diseases, Neurological diseases, etc.	Developed by the National Library of Medicine. Openness:Open Access. License:ODbL 1.0 license.
PadChest [196]	X-ray	MRG	160K images	Pneumonia, Pulmonary edema, Lung cancer, etc.	Published by San Juan Hospital, Spain. Openness:Open Access. License:MIT.
Slake [156]	CT, MRI, X-ray	MVQA	13K image-text pairs	Atelectasis, Consolidation, Pneumothorax, etc.	Image sources include Medical Segmentation Decathlon. Openness:Open Access. License:MIT.
PathVQA [201]	Pathology	MVQA	33K image-text pairs	Lymphoma, Colorectal cancer, Prostate cancer, etc.	Published by Johns Hopkins University. Openness:Open Access. License:MIT.
RadQA [197]	CT, MRI, X-ray	MVQA	50K image-text pairs	Pulmonary Nodule, Liver Cyst, Cerebral Hemorrhage, etc.	Published by National Institutes of Health. Openness:Open Access License:PhysioNet Credentialed Health Data License 1.5.0.
VQA-Med-2019 [198]	X-ray, CT, MRI	MVQA	15K image-text pairs	Pneumonia, Fracture, Neoplasms, etc.	Published by ImageCLEF. Openness:Open Access. License:MIT.
VQA-RAD [199]	CT, MRI, X-ray	MVQA	3.5K image-text pairs	Lung infections, Fractures, Tumors, etc.	Data from MedPix. Openness:Open Access. License:CC0 1.0 Universal.
PubMedVision [200]	X-ray, CT, MRI, Pathology	MVQA	1.2M image-text pairs	Lymphoma, Colorectal cancer, Pneumonia, etc.	Constructed by CUHK. Openness:Open Access License:MIT.
MedTrinity-25M [202]	CT, X-ray, MRI, US, Pathology	Lang-SEG	25M images	Lung cancer, Coronary Stenosis, Alzheimer's disease, etc.	Published by HUST. Openness:Open Access. License:Attribution-NonCommercial 4.0 International.
MosMedData [203, 204]	CT	Lang-SEG	1.1K images	COVID-19	Published by Moscow Center. Openness:Apply for Access. License:MIT.
RefHL-Seg [164]	MRI, CT	Lang-SEG	3K images	Liver Tumors, Liver Cysts, etc.	Published by CUHK. Openness:Open Access. License:MIT.
QaTa-COV19 [205]	CT, X-ray	Lang-SEG	188 images	COVID-19	Published by Alibaba Cloud Tianchi. Openness:Open Access. License:CC BY-NC-SA 3.0 IGO.
PMC-OA [208]	X-ray, CT, MRI, Pathology	Pretrain	1.65M image-text pairs	Pneumonia, Pneumothorax, Tumor, etc.	Published by Shanghai Jiao Tong University. Openness:Open Access. License:MIT.
ROCO [206]	CT, X-ray, MRI	Pretrain	80k image-text pairs	Thorax, Abdomen, Bones, etc.	Published by FHDO. Openness:Open Access. License:MIT.
MedICaT [207]	X-ray, CT, MRI, Pathology	Pretrain	217K image-text pairs	Tumors, Inflammation, Structural abnormalities of organs, etc.	Published by Allen Institute for AI. Openness:Open Access. License:MIT.
PMC-VQA [161]	X-ray, CT, MRI, Pathology	Pretrain	1M image-text pairs	Tumors, Infectious diseases, Cardiovascular diseases, etc.	Published by Shanghai Jiao Tong University. Openness:Open Access. License:MIT.

6.3. Benchmark-based Evaluation

In the evaluation methods of knowledge fusion for medical vision-language models, apart from the above-mentioned quantitative and qualitative evaluations, benchmark-based evaluation is also very common, which includes designing and proposing a new benchmark or making targeted modifications to existing benchmarkss [134, 218–224]. The principles for designing and modifying benchmarks cover aspects such as credibility, fairness, privacy security, robustness, knowledge conflict issues, and data leakage handling.

Credibility emphasizes whether the evaluation model can accurately integrate and apply medical knowledge during the knowledge fusion process, avoiding the generation of incorrect or misleading results. Fairness emphasizes ensuring that the model does not produce biases towards certain specific groups or individuals when dealing with different types of medical data and tasks. Privacy security emphasizes paying attention to whether the model will leak patients' privacy information during the knowledge fusion process and whether it will pose potential threats to patients' health and safety. Robustness emphasizes examining the stability and reliability of the model when facing noise, abnormal data, or malicious attacks [218]. Knowledge conflict issues emphasize identifying and resolving possible knowledge conflict issues that may arise during the knowledge fusion process of the model, ensuring that the model can correctly handle medical knowledge from different sources and types [219]. Data leakage handling emphasizes that in the evaluation of medical vision-language models, handling data leakage is of crucial importance. The dataset needs to be carefully processed to avoid the overlap between the test set and the training set, and a combination of multiple independent datasets can be adopted [220].

Currently, there are several newly proposed benchmarks for the evaluation of knowledge fusion in medical vision-language

models. CARES [221] is a benchmark designed to comprehensively assess the trustworthiness of medical vision-language models. It covers 16 medical image modalities and 27 anatomical structures, conducting a thorough evaluation of the model's ability to integrate knowledge by designing various tasks and assessment metrics. MultiMedEval [222] is an open-source toolkit for evaluating medical vision-language models, encompassing six tasks: image classification, question answering, visual question answering, report summarization, report generation, and natural language inference. It evaluates models across 23 datasets, offering a comprehensive assessment of their multimodal task-handling capabilities and knowledge integration performance. BenchX [223] is a unified benchmarking framework designed for systematic evaluation and comparison of medical vision-language pretraining methods using publicly available chest X-ray datasets. It encompasses nine datasets and four medical tasks, standardized data preprocessing, training-test splits, parameter selection, and unified fine-tuning protocols, applicable to classification, segmentation, and report generation tasks.

7. Future Directions

With the development of artificial intelligence and large models, the knowledge fusion in medical vision-language models faces numerous challenges, which also represent the future research directions. In this section, we will mainly discuss aspects such as data and knowledge, model architecture, evaluation, and other applications.

7.1. Data and Knowledge Aspects

In the past, the collection of knowledge heavily relied on doctor annotations, requiring a significant investment of human resources. Large language models have greatly improved the efficiency of text data processing and can reduce the difficulty of organizing knowledge-based data. Constructing knowledge graphs based on large language models is also a popular research direction currently. Arsenyan et al. [225] demonstrated that LLMs can extract standardized knowledge triples from unstructured medical texts through automated semantic parsing and associate them with multimodal data such as clinical images and lab reports, thereby improving data richness and accessibility. However, the terminological ambiguity of medical texts and the need for privacy protection remain core bottlenecks. Directly constructing data based on large language models may introduce additional hallucinations and reduce the reliability of the generated knowledge content. To mitigate these risks, Huang et al. [226] recommended developing a hybrid framework combining LLM-generated knowledge with rule-based or ontology-guided validation layers, which can serve as a secondary knowledge verification mechanism to ensure the factuality and consistency of extracted triples. Future research could focus on designing robust evaluation metrics and scalable pipelines for integrating human-in-the-loop verification.

In addition, the rapid iteration of medical knowledge requires data to support dynamic updates. The already constructed knowledge needs to be continuously and dynamically expanded and updated in a timely manner. With the continuous release of new clinical research results, drug development progress, and diagnosis and treatment guidelines, the information in medical knowledge graphs must be able to respond quickly to these changes to ensure its accuracy and practicality. Luo et al. [227] proposed an architecture for a real-time monitoring and update system, which leverages natural language processing tools to scan medical literature repositories and clinical trial databases. However, implementation challenges remain, such as how to detect conflicting updates and maintain consistency across versions. A promising direction is the use of change-tracking ontologies and provenance modeling to manage dynamic evolution.

With the popularization of the multimodal capabilities of models, unimodal knowledge bases can hardly meet the requirements of models. Therefore, constructing a multimodal knowledge base has become an inevitable trend in future development [228]. A multimodal knowledge base not only contains text data but also integrates various types of information such as medical images, physiological signals, and genetic data, which can provide more comprehensive and richer knowledge support for medical research and clinical decision-making. Some work has already been done in the direction of multimodal knowledge graphs. For instance, Wang et al. [229] focuses on architectural design for constructing multimodal knowledge graphs, incorporating data preprocessing, feature extraction, and alignment across modalities. However, research on effective fusion strategies—particularly how to semantically align and reason across modalities—is still in its infancy. Potential research directions include using contrastive learning or graph neural networks to bridge the gap between heterogeneous representations. At the same time, the acquisition and processing costs of multimodal data are relatively high, and the quality and integrity of data from different modalities may also vary. Future work should also explore cost-efficient data acquisition methods, such as federated learning frameworks or synthetic data generation, to address the high expense and privacy concerns associated with real multimodal datasets. By solving key problems such as data fusion, semantic alignment, and dynamic updates, multimodal knowledge bases will provide more powerful support for medical research and clinical practice, driving the medical field towards a more accurate direction.

7.2. Model Architecture Aspects

Traditional knowledge fusion methods focus on designing a reasonable model architecture to achieve knowledge fusion at the feature level. These methods have enhanced the model's ability to understand and apply complex knowledge. However, the decision-making process is difficult to understand, and it lacks interpretability, which poses a significant barrier to deploying such models in real-world clinical settings where transparency and accountability are critical [230]. This is particularly crucial in the medical field because medical decisions require clear justifications and explanations. Hossain et al. [231] pointed out that although large language models have improved the model's generation ability to some extent, their complex internal structures have made the interpretability problem more

prominent. To overcome this challenge, recent studies have explored techniques such as attention visualization, concept attribution, and counterfactual reasoning to enhance interpretability in medical applications, enabling clinicians to better understand and trust model decisions. At the same time, due to the large number of parameters and complex structure of large language models, it is very difficult to achieve knowledge fusion at the feature level. Therefore, future research could explore designing modular architectures with interpretable intermediate representations, which can naturally support external knowledge fusion at the feature level while enhancing transparency.

In addition, although large language models (LLMs) contain rich internal knowledge, this knowledge is often stored implicitly, making it difficult to extract and directly apply to specific medical tasks, such as rare disease diagnosis or patient-specific treatment recommendations. To address this issue, researchers have explored approaches such as prompt tuning, retrieval-augmented generation, and adapter-based fine-tuning to better activate and align the internal knowledge of LLMs with task-specific requirements, thereby achieving multi-capability integration. Rasheed et al. [232] designed a reasonable model architecture that can not only support medical visual question answering but also provide segmentation results of relevant medical entities. Such integration not only improves overall model performance but also facilitates the development of unified frameworks that can handle diverse medical tasks—e.g., combining visual report generation, diagnostic reasoning, and entity segmentation in a single system—thus improving clinical workflow efficiency.

Zuo et al. [233] mentioned that a feasible method of knowledge fusion at the model level is the mixture-of-experts (MoE) model, which can address the problem of balancing computational efficiency and performance of large models. In the medical field, different tasks and data types have different requirements for the capabilities of the model. The mixture-of-experts model can dynamically select appropriate expert modules according to the task requirements, thus achieving more efficient computation and more accurate prediction. However, current applications of the mixture-of-experts (MoE) paradigm in the medical domain remain limited. Future research could investigate task-aware expert assignment, interpretable gating mechanisms, and lightweight expert modules tailored to specific clinical subdomains, such as radiology or oncology. For example, Lin et al. [234] emphasized the importance of designing more effective expert module partitioning strategies and interpretable gating mechanisms to fully leverage the advantages of MoE models in clinical decision-making scenarios. Overall, addressing the challenges of interpretability, implicit knowledge utilization, and efficient expert routing in large-scale medical models will require interdisciplinary efforts that combine advances in model architecture, cognitive science, and domain-specific knowledge representation.

7.3. Evaluation Aspects

Currently, the evaluation of knowledge mainly focuses on the accuracy and completeness of knowledge. However, with the improvement of model capabilities and the complexity of application scenarios, merely focusing on these aspects can no longer meet the requirements. The evaluation of knowledge relevance has become an important direction for future evaluation. In the medical field, Xiao et al. [5] emphasized that the practicality of knowledge not only hinges on its accuracy but also critically on its task relevance; for example, irrelevant but accurate knowledge may mislead clinical decision-making, especially in complex diagnostic scenarios. In the future, it is necessary to develop more refined relevance evaluation indicators and methods. By conducting a quantitative analysis of the semantic matching degree between knowledge and task objectives, the knowledge that is truly useful for the task can be screened out.

The correctness of knowledge has always been the core content of evaluation. However, with the diversification and complexity of knowledge sources, the difficulty of correctness evaluation is also increasing. Although large language models can generate a large amount of knowledge, the knowledge may have hallucinations or biases, which brings new challenges to the correctness evaluation of knowledge. Ibrahim et al. [235] proposed that future correctness evaluation systems should integrate expert review, data consistency checks, and automated logical inference. For instance, integrating knowledge-tracing algorithms with expert-curated datasets can help detect subtle hallucinations generated by large language models. At the same time, with the dynamic update of the knowledge graph, the correctness evaluation also needs to be dynamic and real-time, being able to detect and correct errors in knowledge in a timely manner.

In addition, the logical coherence of knowledge is one of the important metrics for evaluating the quality of knowledge. [236] In the medical field, the logical coherence of knowledge not only affects the understanding and reasoning ability of the model but also directly relates to the reliability of clinical decisions. For example, the causal relationships and pathological processes in medical knowledge all need to have strict logic. In the future, it is necessary to develop more advanced logical coherence evaluation tools. For instance, some work suggested that using path analysis and causal reasoning in knowledge graphs can help identify contradictions. Future work could explore embedding-based logical constraint detection to automatically pinpoint incoherent relations, especially in rapidly evolving medical domains.

7.4. Other Application Aspects

In recent years, challenges such as the difficulty of timely knowledge updates in dynamic domains and growing concerns over model security have spurred active research in knowledge editing technology [237], which aims to address these limitations by providing adaptable and secure mechanisms for modifying the knowledge embedded in large language models. Knowledge editing is a technology aimed at dynamically correcting and updating the knowledge base of large language models. It enables the addition, deletion, modification, and query of specific knowledge without completely retraining the model. This technology is particularly important in medical large language models because knowledge in the medical field

updates rapidly and has extremely high requirements for accuracy. Pan et al. [238] pointed out that through parameter-efficient fine-tuning [239] or incremental learning [240], allow new medical knowledge to be injected into models without catastrophic forgetting. However, future research should explore hybrid frameworks that combine these approaches with real-time validation systems, ensuring that updates are both accurate and clinically safe. Currently, for medical vision-language models, achieving targeted knowledge updates of large models through knowledge editing is a future research direction.

The model security issues brought about by the fusion of knowledge have often been overlooked in the past. However, with the widespread application of models in the medical field, this issue has become increasingly prominent. The privacy and sensitivity of medical data require that models must have a high level of security when fusing knowledge. For example, a model may leak patients' personal information or medical data during the process of knowledge fusion, which will pose a serious threat to patients' privacy and security. In the future, it is necessary to strengthen the security design of models. The privacy and security of medical data can be guaranteed through encryption technology, differential privacy and other methods [241]. At the same time, a strict model security assessment mechanism also needs to be established to conduct real-time monitoring and assessment of the security risks of models during the knowledge fusion process, ensuring the safe and reliable operation of the models.

From a longer-term perspective, knowledge fusion technology is expected to become one of the core drivers for the continuous evolution of medical vision-language models. Structured medical knowledge, as an external memory mechanism, can help the model achieve continuous learning and reduce the problem of forgetting old knowledge when updating new data [242]. Secondly, in actual deployment scenarios across hospitals, regions, and even languages, the model faces the problem of data distribution offset. The introduction of a knowledge fusion mechanism can provide stable semantic information to a certain extent, and improve the model's domain adaptability and robustness for lifelong learning, which has been proven in other fields [243].

Finally, improving the knowledge-based reasoning ability is one of the important goals for the future development of models. In the past, models often relied on simple rules or shallow correlation analysis during the reasoning process, making it difficult to handle complex medical problems. The emergence of large language models has provided more powerful language understanding and generation capabilities for knowledge-based reasoning. However, how to make full use of these capabilities to achieve more accurate and efficient reasoning remains a challenge. Existing methods mainly rely on designing long chain of thought to improve the self-reflection ability of models during generation. In the future, more advanced reasoning algorithms and frameworks need to be developed. Plaat et al. [244] proposed that integrating structured knowledge from knowledge graphs with the implicit representations of large language models can enhance reasoning. To tackle real-world medical challenges, future work should focus on building hybrid reasoning pipelines that incorporate causal inference, symbolic logic, and clinical decision trees, thereby enabling models to handle complex, high-stakes diagnostic tasks with greater reliability.

8. Conclusion

In this paper, we present a comprehensive review of knowledge fusion in medical vision-language deep learning models. By synthesizing a wide range of frontier studies, we systematically elucidate the foundations of medical knowledge fusion, classify key types of medical knowledge, and analyze its diverse applications across various medical vision-language tasks. We explore deep learning-based fusion algorithms and summarize the four progressive stages of knowledge fusion, offering a structured understanding of current methodologies and their intricacies.

To deepen insight into future directions, we further identify several specific technical challenges and emerging trends. First, multi-modal alignment remains a core difficulty, especially when attempting fine-grained alignment between unstructured medical text and heterogeneous visual modalities such as radiology and pathology images. Second, domain-specific knowledge grounding—particularly integrating symbolic clinical knowledge from medical ontologies with pre-trained large vision-language models poses new challenges in terms of scalability, representation consistency, and reasoning fidelity. Third, the evaluation of knowledge fusion efficacy in clinical tasks is still immature; current benchmarks fail to fully capture the interpretability and knowledge-awareness of models. Fourth, with the rapid development of large-scale vision-language foundation models and GPT-style medical LLMs, how to achieve continual learning and personalized adaptation in knowledge fusion while maintaining patient privacy and fairness is an urgent open issue. Finally, prompt-based fusion strategies and retrieval-augmented generation (RAG) pipelines represent a promising research direction, allowing models to dynamically incorporate external knowledge into both understanding and generation tasks.

By actively tackling these challenges, future research will not only enhance model robustness and interpretability, but also support safer and more accurate clinical decision-making. We hope this review provides a clear, structured understanding of current progress and inspires further exploration in knowledge fusion for medical vision-language models.

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References

[1] M. Rehman, I. Shafi, J. Ahmad, C. O. Garcia, A. E. P. Barrera, I. Ashraf, Advancement in medical report generation: current practices, challenges, and future direc-

tions, Medical & Biological Engineering & Computing (2024) 1–22.

[2] Z. Lin, D. Zhang, Q. Tao, D. Shi, G. Haffari, Q. Wu, M. He, Z. Ge, Medical visual question answering: A survey, Artificial Intelligence in Medicine 143 (2023) 102611.

[3] T. Zhou, F. Zhang, B. Chang, W. Wang, Y. Yuan, E. Konukoglu, D. Cremers, Image segmentation in foundation model era: A survey, arXiv preprint arXiv:2408.12957 (2024).

[4] P. Shrestha, S. Amgain, B. Khanal, C. A. Linte, B. Bhattarai, Medical vision language pretraining: A survey, arXiv preprint arXiv:2312.06224 (2023).

[5] H. Xiao, F. Zhou, X. Liu, T. Liu, Z. Li, X. Liu, X. Huang, A comprehensive survey of large language models and multimodal large language models in medicine, Information Fusion (2024) 102888.

[6] J. Hu, Y. Li, H. Sun, Y. Song, C. Zhang, L. Lin, Y.-W. Chen, Lga: A language guide adapter for advancing the sam model's capabilities in medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 610–620.

[7] I. Hartsock, G. Rasool, Vision-language models for medical report generation and visual question answering: A review, Frontiers in Artificial Intelligence 7 (2024) 1430984.

[8] B. Yan, M. Pei, Clinical-bert: Vision-language pretraining for radiograph diagnosis and reports generation, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 36, 2022, pp. 2982–2990.

[9] J. Huang, Y. Chen, Y. Li, Z. Yang, X. Gong, F. L. Wang, X. Xu, W. Liu, Medical knowledge-based network for patient-oriented visual question answering, Information Processing & Management 60 (2) (2023) 103241.

[10] Z. Chen, G. Li, X. Wan, Align, reason and learn: Enhancing medical vision-and-language pre-training with knowledge, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022, pp. 5152–5161.

[11] J. Yang, L. T. Yang, H. Wang, Y. Gao, Y. Zhao, X. Xie, Y. Lu, Representation learning for knowledge fusion and reasoning in cyber–physical–social systems: Survey and perspectives, Information Fusion 90 (2023) 59–73.

[12] D. Xu, H. Zhu, Y. Huang, Z. Jin, W. Ding, H. Li, M. Ran, Vision-knowledge fusion model for multi-domain medical report generation, Information Fusion 97 (2023) 101817.

[13] P. Xia, K. Zhu, H. Li, H. Zhu, Y. Li, G. Li, L. Zhang, H. Yao, RULE: reliable multimodal RAG for factuality in medical vision language models, in: EMNLP, Association for Computational Linguistics, 2024, pp. 1081–1093.

[14] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, J. Gao, Llava-med: Training a large language-and-vision assistant for biomedicine in one day, in: NeurIPS, 2023.

[15] X. Guo, W. Chai, S. Li, G. Wang, Llava-ultra: Large chinese language and vision assistant for ultrasound, in: ACM Multimedia, ACM, 2024, pp. 8845–8854.

[16] X. Huang, H. Huang, L. Shen, Y. Yang, F. Shang, J. Liu, J. Liu, A refer-and-ground multimodal large language model for biomedicine, in: MICCAI (12), Vol. 15012 of Lecture Notes in Computer Science, Springer, 2024, pp. 399–409.

[17] K. Zhang, R. Zhou, E. Adhikarla, Z. Yan, Y. Liu, J. Yu, Z. Liu, X. Chen, B. D. Davison, H. Ren, J. Huang, C. Chen, Y. Zhou, S. Fu, W. Liu, T. Liu, X. Li, Y. Chen, L. He, J. Zou, Q. Li, H. Liu, L. Sun, A generalist vision–language foundation model for diverse biomedical tasks, Nature Medicine 30 (11) (2024) 3129–3141.

[18] Y. Khare, V. Bagal, M. Mathew, A. Devi, U. D. Priyakumar, C. Jawahar, Mmbert: Multimodal bert pretraining for improved medical vqa, in: 2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI), IEEE, 2021, pp. 1033–1036.

[19] S.-C. Huang, L. Shen, M. P. Lungren, S. Yeung, Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 3942–3951.

[20] C. Wu, X. Zhang, Y. Zhang, Y. Wang, W. Xie, Medklip: Medical knowledge enhanced language-image pretraining for x-ray diagnosis, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 21372–21383.

[21] H. Liu, D. Wei, D. Lu, J. Sun, L. Wang, Y. Zheng, M3ae: Multimodal representation learning for brain tumor segmentation with missing modalities, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 37, 2023, pp. 1657–1665.

[22] H. Pan, S. He, K. Zhang, B. Qu, C. Chen, K. Shi, Amam: an attention-based multimodal alignment model for medical visual question answering, Knowledge-Based Systems 255 (2022) 109763.

[23] M. Guo, H. Yi, Z. Qin, H. Wang, A. Men, Q. Lao, Multiple prompt fusion for zero-shot lesion detection using vision-language models, in: International Conference on

Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 283–292.

[24] P. Xia, K. Zhu, H. Li, T. Wang, W. Shi, S. Wang, L. Zhang, J. Zou, H. Yao, Mmed-rag: Versatile multimodal rag system for medical vision language models (2024). arXiv:2410.13085.

[25] X. Qu, Q. Chen, W. Wei, J. Sun, J. Dong, Alleviating hallucination in large vision-language models with active retrieval augmentation (2024). arXiv:2408.00555.

[26] W. Ding, H. Wang, J. Huang, H. Ju, Y. Geng, C.-T. Lin, W. Pedrycz, Ftranscnn: Fusing transformer and a cnn based on fuzzy logic for uncertain medical image segmentation, Information Fusion 99 (2023) 101880.

[27] A. Zafar, K. Mishra, A. Ekbal, Medex: Enhancing medical question-answering with first-order logic based reasoning and knowledge injection, in: Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 9701–9720.

[28] Z. Qin, H. Yi, Q. Lao, K. Li, Medical image understanding with pretrained vision language models: A comprehensive study, arXiv preprint arXiv:2209.15517 (2022).

[29] K. Sun, S. Xue, F. Sun, H. Sun, Y. Luo, L. Wang, S. Wang, N. Guo, L. Liu, T. Zhao, et al., Medical multimodal foundation models in clinical diagnosis and treatment: Applications, challenges, and future directions, arXiv preprint arXiv:2412.02621 (2024).

[30] O. WHO, International classification of diseases, WHO [Internet] (1992).

[31] C. E. Lipscomb, Medical subject headings (mesh), Bulletin of the Medical Library Association 88 (3) (2000) 265.

[32] T. B. Üstün, S. Chatterji, J. Bickenbach, N. Kostanjsek, M. Schneider, The international classification of functioning, disability and health: a new tool for understanding disability and health, Disability and rehabilitation 25 (11-12) (2003) 565–571.

[33] G. Nahler, G. Nahler, Anatomical therapeutic chemical classification system (atc), Dictionary of pharmaceutical medicine (2009) 8–8.

[34] E. G. Brown, L. Wood, S. Wood, The medical dictionary for regulatory activities (meddra), Drug safety 20 (2) (1999) 109–117.

[35] A. Davitkova, D. Gjurovski, S. Michel, Lmkg: learned models for cardinality estimation in knowledge graphs, arXiv preprint arXiv:2102.10588 (2021).

[36] A. Santos, A. R. Colaço, A. B. Nielsen, L. Niu, P. E. Geyer, F. Coscia, N. J. W. Albrechtsen, F. Mundt, L. J. Jensen, M. Mann, Clinical knowledge graph integrates proteomics data into clinical decision-making, bioRxiv (2020) 2020–05.

[37] J.-B. Delbrouck, P. Chambon, Z. Chen, M. Varma, A. Johnston, L. Blankemeier, D. Van Veen, T. Bui, S. Truong, C. Langlotz, RadGraph-XL: A large-scale expert-annotated dataset for entity and relation extraction from radiology reports, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Findings of the Association for Computational Linguistics ACL 2024, Association for Computational Linguistics, Bangkok, Thailand and virtual meeting, 2024, pp. 12902–12915.

[38] A. E. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C.-y. Deng, R. G. Mark, S. Horng, Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports, Scientific data 6 (1) (2019) 317.

[39] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, et al., Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 33, 2019, pp. 590–597.

[40] P. Chandak, K. Huang, M. Zitnik, Building a knowledge graph to enable precision medicine, Nature Scientific Data (2023).

[41] A. E. Johnson, L. Bulgarelli, L. Shen, A. Gayles, A. Shammout, S. Horng, T. J. Pollard, S. Hao, B. Moody, B. Gow, et al., Mimic-iv, a freely accessible electronic health record dataset, Scientific data 10 (1) (2023) 1.

[42] A. E. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. Anthony Celi, R. G. Mark, Mimic-iii, a freely accessible critical care database, Scientific data 3 (1) (2016) 1–9.

[43] T. J. Pollard, A. E. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, O. Badawi, The eicu collaborative research database, a freely available multi-center database for critical care research, Scientific data 5 (1) (2018) 1–13.

[44] M. Wornow, R. Thapa, E. Steinberg, J. Fries, N. Shah, Ehrshot: An ehr benchmark for few-shot evaluation of foundation models, Advances in Neural Information Processing Systems 36 (2023) 67125–67137.

[45] P. J. Thoral, J. M. Peppink, R. H. Driessen, E. J. Sijbrands, E. J. Kompanje, L. Kaplan, H. Bailey, J. Kesecioglu, M. Cecconi, M. Churpek, et al., Sharing icu patient data responsibly under the society of critical care medicine/european society of intensive care medicine joint data science collaboration: the amsterdam university medical centers database (amsterdamumcdb) example, Critical care medicine 49 (6) (2021) e563–e577.

[46] X. Yue, X. Huang, Z. Xu, Y. Chen, C. Xu, Involving logical clinical knowledge into deep neural networks to improve bladder tumor segmentation, Medical Image Analysis 95 (2024) 103189.

[47] D. Sarvamangala, R. V. Kulkarni, Convolutional neural networks in medical image understanding: a survey, Evolutionary intelligence 15 (1) (2022) 1–22.

[48] F. Krones, U. Marikkar, G. Parsons, A. Szmul, A. Mahdi, Review of multimodal machine learning approaches in healthcare, Information Fusion 114 (2025) 102690.

[49] A. Krizhevsky, I. Sutskever, G. E. Hinton, Imagenet classification with deep convolutional neural networks, Advances in neural information processing systems 25 (2012).

[50] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint arXiv:1409.1556 (2014).

[51] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, A. Rabinovich, Going deeper with convolutions, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2015, pp. 1–9.

[52] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 770–778.

[53] G. Huang, Z. Liu, L. Van Der Maaten, K. Q. Weinberger, Densely connected convolutional networks, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2017, pp. 4700–4708.

[54] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18, Springer, 2015, pp. 234–241.

[55] D. Tran, L. Bourdev, R. Fergus, L. Torresani, M. Paluri, Learning spatiotemporal features with 3d convolutional networks, in: Proceedings of the IEEE international conference on computer vision, 2015, pp. 4489–4497.

[56] K. Kamnitsas, C. Ledig, V. F. Newcombe, J. P. Simpson, A. D. Kane, D. K. Menon, D. Rueckert, B. Glocker, Efficient multi-scale 3d cnn with fully connected crf for accurate brain lesion segmentation, Medical image analysis 36 (2017) 61–78.

[57] J. L. Elman, Finding structure in time, Cognitive science 14 (2) (1990) 179–211.

[58] A. Graves, A. Graves, Long short-term memory, Supervised sequence labelling with recurrent neural networks (2012) 37–45.

[59] K. Cho, B. van Merrienboer, C. Gulcehre, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using rnn encoder-decoder for statistical machine translation, in: Conference on Empirical Methods in Natural Language Processing (EMNLP 2014), 2014.

[60] L. Li, L. Jin, Z. Jiang, D. Song, D. Huang, Biomedical named entity recognition based on extended recurrent neural networks, in: 2015 IEEE International Conference on bioinformatics and biomedicine (BIBM), IEEE, 2015, pp. 649–652.

[61] A. Gasimova, Automated enriched medical concept generation for chest x-ray images, in: Interpretability of machine intelligence in medical image computing and multimodal learning for clinical decision support, Springer, 2019, pp. 83–92.

[62] M. Gu, X. Huang, Y. Fang, Automatic generation of pulmonary radiology reports with semantic tags, in: 2019 IEEE 11th International Conference on Advanced Infocomm Technology (ICAIT), IEEE, 2019, pp. 162–167.

[63] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Advances in neural information processing systems 30 (2017).

[64] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers), 2019, pp. 4171–4186.

[65] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever, et al., Improving language understanding by generative pre-training (2018).

[66] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, in: International Conference on Learning Representations.

[67] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF international conference on computer vision, 2021, pp. 10012–10022.

[68] Y. Xiao, Q. Yuan, K. Jiang, J. He, C.-W. Lin, L. Zhang, Ttst: A top-k token selective transformer for remote sensing image super-resolution, IEEE Transactions on Image Processing 33 (2024) 738–752.

[69] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, Bioinformatics 36 (4) (2020) 1234–1240.

[70] K. Huang, J. Altosaar, R. Ranganath, Clinicalbert: Modeling clinical notes and predicting hospital readmission, arXiv preprint arXiv:1904.05342 (2019).

[71] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, Y. Zhou, Transunet: Transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306 (2021).

[72] W. Wenxuan, C. Chen, D. Meng, Y. Hong, Z. Sen, L. Jiangyun, Transbts: Multimodal brain tumor segmentation using transformer, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2021, pp. 109–119.

[73] O. Dalmaz, M. Yurt, T. Çukur, Resvit: residual vision transformers for multimodal medical image synthesis, IEEE Transactions on Medical Imaging 41 (10) (2022) 2598–2614.

[74] Z. Li, Y. Jiang, M. Lu, R. Li, Y. Xia, Survival prediction via hierarchical multimodal co-attention transformer: A computational histology-radiology solution, IEEE Transactions on Medical Imaging 42 (9) (2023) 2678–2689.

[75] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: International Conference on Learning Representations, 2017.

[76] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, Graph attention networks, in: International Conference on Learning Representations, 2018.

[77] Q. Huang, G. Li, Knowledge graph based reasoning in medical image analysis: A scoping review, Computers in Biology and Medicine 182 (2024) 109100.

[78] J. Zhang, D. Xu, Y. Lou, Y. Huang, A novel multi-atlas fusion model based on contrastive learning for functional connectivity graph diagnosis, in: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2024, pp. 2295–2299.

[79] R. Bhalodia, A. Hatamizadeh, L. Tam, Z. Xu, X. Wang, E. Turkbey, D. Xu, Improving pneumonia localization via cross-attention on medical images and reports, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part II 24, Springer, 2021, pp. 571–581.

[80] H. Gong, G. Chen, S. Liu, Y. Yu, G. Li, Cross-modal self-attention with multi-task pre-training for medical visual question answering, in: Proceedings of the 2021 international conference on multimedia retrieval, 2021, pp. 456–460.

[81] D. Xu, Y. Chen, J. Zhang, Y. Lou, H. Wang, J. He, Y. Huang, Radiology report generation via structured knowledge-enhanced multi-modal attention and contrastive learning, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2023, pp. 2320–2325.

[82] S. Yang, X. Wu, S. Ge, S. K. Zhou, L. Xiao, Knowledge matters: Chest radiology report generation with general and specific knowledge, Medical image analysis 80 (2022) 102510.

[83] X. Huang, H. Gong, A dual-attention learning network with word and sentence embedding for medical visual question answering, IEEE Transactions on Medical Imaging 43 (2) (2023) 832–845.

[84] Z. Huang, X. Zhang, S. Zhang, Kiut: Knowledge-injected u-transformer for radiology report generation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 19809–19818.

[85] M. Hassanin, S. Anwar, I. Radwan, F. S. Khan, A. Mian, Visual attention methods in deep learning: An in-depth survey, Information Fusion 108 (2024) 102417.

[86] R. Hou, H. Chang, B. Ma, S. Shan, X. Chen, Cross attention network for few-shot classification, Advances in neural information processing systems 32 (2019).

[87] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, W. Liu, Ccnet: Criss-cross attention for semantic segmentation, in: Proceedings of the IEEE/CVF international conference on computer vision, 2019, pp. 603–612.

[88] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, J. Carreira, Perceiver: General perception with iterative attention, in: International conference on machine learning, PMLR, 2021, pp. 4651–4664.

[89] K.-H. Lee, X. Chen, G. Hua, H. Hu, X. He, Stacked cross attention for image-text matching, in: Proceedings of the European conference on computer vision (ECCV), 2018, pp. 201–216.

[90] Y. Xie, B. Yang, Q. Guan, J. Zhang, Q. Wu, Y. Xia, Attention mechanisms in medical image segmentation: A survey, arXiv preprint arXiv:2305.17937 (2023).

[91] H. Sun, X. Zeng, T. Xu, G. Peng, Y. Ma, Computer-aided diagnosis in histopathological images of the endometrium using a convolutional neural network and attention mechanisms, IEEE journal of biomedical and health informatics 24 (6) (2019) 1664–1676.

[92] W. Zhao, R. Wang, Y. Qi, M. Lou, Y. Wang, Y. Yang, X. Deng, Y. Ma, Bascnet: Bilateral adaptive spatial and channel attention network for breast density classification in the mammogram, Biomedical Signal Processing and Control 70 (2021) 103073.

[93] S. Woo, J. Park, J.-Y. Lee, I. S. Kweon, Cbam: Convolutional block attention module, in: Proceedings of the European conference on computer vision (ECCV), 2018, pp. 3–19.

[94] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 16000–16009.

[95] P. Li, J. Gu, J. Kuen, V. I. Morariu, H. Zhao, R. Jain, V. Manjunatha, H. Liu, Selfdoc: Self-supervised document representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 5652–5660.

[96] C. Sun, A. Myers, C. Vondrick, K. Murphy, C. Schmid, Videobert: A joint model for video and language representation learning, in: Proceedings of the IEEE/CVF international conference on computer vision, 2019, pp. 7464–7473.

[97] H. Bao, W. Wang, L. Dong, F. Wei, Vl-beit: Generative vision-language pretraining, arXiv preprint arXiv:2206.01127 (2022).

[98] H.-Y. Zhou, C. Lian, L. Wang, Y. Yu, Advancing radiograph representation learning with masked record modeling, in: The Eleventh International Conference on Learning Representations.

[99] M.-I. Georgescu, Masked autoencoders for unsupervised anomaly detection in medical images, Procedia Computer Science 225 (2023) 969–978.

[100] Z. Chen, Y. Du, J. Hu, Y. Liu, G. Li, X. Wan, T.-H. Chang, Multi-modal masked autoencoders for medical vision-and-language pre-training, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2022, pp. 679–689.

[101] J. H. Moon, H. Lee, W. Shin, Y.-H. Kim, E. Choi, Multi-modal understanding and generation for medical images and text via vision-language pre-training, IEEE Journal of Biomedical and Health Informatics 26 (12) (2022) 6070–6080.

[102] A. v. d. Oord, Y. Li, O. Vinyals, Representation learning with contrastive predictive coding, arXiv preprint arXiv:1807.03748 (2018).

[103] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2020, pp. 9729–9738.

[104] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International conference on machine learning, PmLR, 2021, pp. 8748–8763.

[105] H. Akbari, L. Yuan, R. Qian, W.-H. Chuang, S.-F. Chang, Y. Cui, B. Gong, Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text, Advances in neural information processing systems 34 (2021) 24206–24221.

[106] A. Guzhov, F. Raue, J. Hees, A. Dengel, Audioclip: Extending clip to image, text and audio, in: ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2022, pp. 976–980.

[107] Y. Xiao, Q. Yuan, K. Jiang, J. He, Y. Wang, L. Zhang, From degrade to upgrade: Learning a self-supervised degradation guided adaptive network for blind remote sensing image super-resolution, Information Fusion 96 (2023) 297–311.

[108] Z. Wang, Z. Wu, D. Agarwal, J. Sun, Medclip: Contrastive learning from unpaired medical images and text, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing, Vol. 2022, 2022, p. 3876.

[109] H. Sowrirajan, J. Yang, A. Y. Ng, P. Rajpurkar, Moco pretraining improves representation and transferability of chest x-ray models, in: Medical Imaging with Deep Learning, PMLR, 2021, pp. 728–744.

[110] M. Y. Lu, R. J. Chen, F. Mahmood, Semi-supervised breast cancer histology classification using deep multiple instance learning and contrast predictive coding (conference presentation), in: Medical imaging 2020: digital pathology, Vol. 11320, SPIE, 2020, p. 113200J.

[111] J. Silva-Rodriguez, H. Chakor, R. Kobbi, J. Dolz, I. B. Ayed, A foundation language-image model of the retina (flair): Encoding expert knowledge in text supervision, Medical Image Analysis 99 (2025) 103357.

[112] Y.-C. Chen, L. Li, L. Yu, A. El Kholy, F. Ahmed, Z. Gan, Y. Cheng, J. Liu, Uniter: Universal image-text representation learning, in: European conference on computer vision, Springer, 2020, pp. 104–120.

[113] J. Li, D. Li, S. Savarese, S. Hoi, Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models, in: International conference on machine learning, PMLR, 2023, pp. 19730–19742.

[114] J. Lu, D. Batra, D. Parikh, S. Lee, Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks, Advances in neural information processing systems 32 (2019).

[115] Z. Wang, J. Yu, A. W. Yu, Z. Dai, Y. Tsvetkov, Y. Cao, Simvlm: Simple visual language model pretraining with weak supervision, in: International Conference on Learning Representations.

[116] W. Li, C. Gao, G. Niu, X. Xiao, H. Liu, J. Liu, H. Wu, H. Wang, Unimo: Towards unified-modal understanding and generation via cross-modal contrastive learning, in: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), 2021, pp. 2592–2607.

[117] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, et al., Language models are unsupervised multitask learners, OpenAI blog 1 (8) (2019) 9.

[118] C. Li, C. Wong, S. Zhang, N. Usuyama, H. Liu, J. Yang, T. Naumann, H. Poon, J. Gao, Llava-med: Training a large language-and-vision assistant for biomedicine in one day, Advances in Neural Information Processing Systems 36 (2023) 28541–28564.

[119] H. Zhang, J. Chen, F. Jiang, F. Yu, Z. Chen, G. H. Chen, J. Li, X. Wu, Z. Zhiyi, Q. Xiao, et al., Huatuogpt, towards taming language model to be a doctor, in: The 2023 Conference on Empirical Methods in Natural Language Processing.

[120] G. Hinton, O. Vinyals, J. Dean, Distilling the knowledge in a neural network, arXiv preprint arXiv:1503.02531 (2015).

[121] Q. Dou, Q. Liu, P. A. Heng, B. Glocker, Unpaired multimodal segmentation via knowledge distillation, IEEE transactions on medical imaging 39 (7) (2020) 2415–2425.

[122] X. Xing, Z. Chen, Y. Hou, Y. Yuan, Gradient modulated contrastive distillation of low-rank multi-modal knowledge for disease diagnosis, Medical Image Analysis 88 (2023) 102874.

[123] J. Wang, S. Huang, H. Du, Y. Qin, H. Wang, W. Zhang, Mhkd-mvqa: Multimodal hierarchical knowledge distillation for medical visual question answering, in: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2022, pp. 567–574.

[124] Q. Lin, R. Mao, J. Liu, F. Xu, E. Cambria, Fusing topology contexts and logical rules in language models for knowledge graph completion, Information Fusion 90 (2023) 253–264.

[125] X. Zhao, Y. Jia, A. Li, R. Jiang, Y. Song, Multi-source knowledge fusion: a survey, World Wide Web 23 (2020) 2567–2592.

[126] X. L. Dong, F. Naumann, Data fusion: resolving data conflicts for integration, Proceedings of the VLDB Endowment 2 (2) (2009) 1654–1655.

[127] A. P. Dempster, Upper and lower probabilities induced by a multivalued mapping, in: Classic works of the Dempster-Shafer theory of belief functions, Springer, 2008, pp. 57–72.

[128] N. Lao, T. Mitchell, W. Cohen, Random walk inference and learning in a large scale knowledge base, in: Proceedings of the 2011 conference on empirical methods in natural language processing, 2011, pp. 529–539.

[129] X. L. Dong, E. Gabrilovich, K. Murphy, V. Dang, W. Horn, C. Lugaresi, S. Sun, W. Zhang, Knowledge-based trust: Estimating the trustworthiness of web sources, Proceedings of the VLDB Endowment 8 (9) (2015).

[130] B. Zhao, J. Han, A probabilistic model for estimating real-valued truth from conflicting sources, Proc. of QDB 1817 (2012).

[131] J. Hao, L. Li, X. Qin, Biomedical causal relation extraction via data augmentation and multi-source knowledge fusion, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2023, pp. 1112–1117.

[132] M. Wei, L. Wang, Y. Li, Z. Li, B. Zhao, X. Su, Y. Wei, Z. You, Biokg-cmi: a multi-source feature fusion model based on biological knowledge graph for predicting circrna-mirna interactions, Science China Information Sciences 67 (8) (2024) 189104.

[133] M. Li, W. Cai, K. Verspoor, S. Pan, X. Liang, X. Chang, Cross-modal clinical graph transformer for ophthalmic report generation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 20656–20665.

[134] D. Xu, Y. Chen, J. Wang, Y. Huang, H. Wang, Z. Jin, H. Wang, W. Yue, J. He, H. Li, Y. Huang, Mlevlm: Improve multi-level progressive capabilities based on multimodal large language model for medical visual question answering, in: ACL (Findings), Association for Computational Linguistics, 2024, pp. 4977–4997.

[135] J. Chen, C. Gui, R. Ouyang, A. Gao, S. Chen, G. Chen, X. Wang, Z. Cai, K. Ji, X. Wan, B. Wang, Towards injecting medical visual knowledge into multimodal llms at scale, in: EMNLP, Association for Computational Linguistics, 2024, pp. 7346–7370.

[136] Y. Li, Q. Yang, F. L. Wang, L.-K. Lee, Y. Qu, T. Hao, Asymmetric cross-modal attention network with multimodal augmented mixup for medical visual question answering, Artificial Intelligence in Medicine 144 (2023) 102667.

[137] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, Advances in neural information processing systems 26 (2013).

[138] C. Dou, Y. Zhang, Z. Jin, W. Jiao, H. Zhao, Y. Zhao, Z. Tao, Integrating physician diagnostic logic into large language models: Preference learning from process

feedback, in: Findings of the Association for Computational Linguistics ACL 2024, 2024, pp. 2453–2473.

[139] Z. Yuan, Q. Jin, C. Tan, Z. Zhao, H. Yuan, F. Huang, S. Huang, Ramm: Retrieval-augmented biomedical visual question answering with multi-modal pre-training (2023). arXiv:2303.00534.

[140] C. Y. Li, X. Liang, Z. Hu, E. P. Xing, Knowledge-driven encode, retrieve, paraphrase for medical image report generation, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 33, 2019, pp. 6666–6673.

[141] Y. Zhang, X. Wang, Z. Xu, Q. Yu, A. Yuille, D. Xu, When radiology report generation meets knowledge graph, in: Proceedings of the AAAI conference on artificial intelligence, Vol. 34, 2020, pp. 12910–12917.

[142] F. Liu, X. Wu, S. Ge, W. Fan, Y. Zou, Exploring and distilling posterior and prior knowledge for radiology report generation, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2021, pp. 13753–13762.

[143] X. Jia, Y. Xiong, J. Zhang, Y. Zhang, Y. Zhu, S. Y. Philip, Few-shot radiology report generation via knowledge transfer and multi-modal alignment, in: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2022, pp. 1574–1579.

[144] S. Jain, A. Agrawal, A. Saporta, S. Q. Truong, D. N. Duong, T. Bui, P. Chambon, Y. Zhang, M. P. Lungren, A. Y. Ng, et al., Radgraph: Extracting clinical entities and relations from radiology reports, arXiv preprint arXiv:2106.14463 (2021).

[145] G. Liu, Y. Liao, F. Wang, B. Zhang, L. Zhang, X. Liang, X. Wan, S. Li, Z. Li, S. Zhang, et al., Medical-vlbert: Medical visual language bert for covid-19 ct report generation with alternate learning, IEEE Transactions on Neural Networks and Learning Systems 32 (9) (2021) 3786–3797.

[146] S. Yang, X. Wu, S. Ge, Z. Zheng, S. K. Zhou, L. Xiao, Radiology report generation with a learned knowledge base and multi-modal alignment, Medical Image Analysis 86 (2023) 102798.

[147] S. Bu, T. Li, Y. Yang, Z. Dai, Instance-level expert knowledge and aggregate discriminative attention for radiology report generation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 14194–14204.

[148] S. Wu, B. Yang, Z. Ye, H. Wang, H. Zheng, T. Zhang, Improving medical report generation with adapter tuning and knowledge enhancement in vision-language foundation models, arXiv preprint arXiv:2312.03970 (2023).

[149] S. Bu, Y. Song, T. Li, Z. Dai, Dynamic knowledge prompt for chest x-ray report generation, in: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024), 2024, pp. 5425–5436.

[150] T. Do, B. X. Nguyen, E. Tjiputra, M. Tran, Q. D. Tran, A. Nguyen, Multiple meta-model quantifying for medical visual question answering, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part V 24, Springer, 2021, pp. 64–74.

[151] S. Tascon-Morales, P. Márquez-Neila, R. Sznitman, Consistency-preserving visual question answering in medical imaging, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2022, pp. 386–395.

[152] D. Gupta, S. Suman, A. Ekbal, Hierarchical deep multimodal network for medical visual question answering, Expert Systems with Applications 164 (2021) 113993.

[153] T. Kornuta, D. Rajan, C. Shivade, A. Asseman, A. S. Ozcan, Leveraging medical visual question answering with supporting facts, in: Conference and Labs of the Evaluation Forum, CEUR-WS, 2019.

[154] Z. Liao, Q. Wu, C. Shen, A. van den Hengel, J. Verjans, Aiml at vqa-med 2020: Knowledge inference via a skeleton-based sentence mapping approach for medical domain visual question answering., in: CLEF (Working Notes), 2020.

[155] W. Zheng, L. Yan, F.-Y. Wang, C. Gou, Learning from the guidance: Knowledge embedded meta-learning for medical visual question answering, in: Neural Information Processing: 27th International Conference, ICONIP 2020, Bangkok, Thailand, November 18–22, 2020, Proceedings, Part IV 27, Springer, 2020, pp. 194–202.

[156] B. Liu, L.-M. Zhan, L. Xu, L. Ma, Y. Yang, X.-M. Wu, Slake: A semantically-labeled knowledge-enhanced dataset for medical visual question answering, in: 2021 IEEE 18th international symposium on biomedical imaging (ISBI), IEEE, 2021, pp. 1650–1654.

[157] X. Hu, L. Gu, Q. An, M. Zhang, L. Liu, K. Kobayashi, T. Harada, R. M. Summers, Y. Zhu, Expert knowledge-aware image difference graph representation learning for difference-aware medical visual question answering, in: Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023, pp. 4156–4165.

[158] P. Li, G. Liu, J. He, Z. Zhao, S. Zhong, Masked vision and language pre-training with unimodal and multimodal contrastive losses for medical visual question answering,

in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 374–383.

[159] S. He, H. Pan, K. Zhang, C. Gong, Z. Li, A weak supervision-based robust pretraining method for medical visual question answering, in: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2023, pp. 1957–1960.

[160] G. Liu, J. He, P. Li, Z. Zhao, S. Zhong, Cross-modal self-supervised vision language pre-training with multiple objectives for medical visual question answering, Journal of Biomedical Informatics 160 (2024) 104748.

[161] X. Zhang, C. Wu, Z. Zhao, W. Lin, Y. Zhang, Y. Wang, W. Xie, Pmc-vqa: Visual instruction tuning for medical visual question answering, arXiv preprint arXiv:2305.10415 (2023).

[162] Q. Yan, J. Duan, J. Wang, Multi-modal concept alignment pre-training for generative medical visual question answering, in: Findings of the Association for Computational Linguistics ACL 2024, 2024, pp. 5378–5389.

[163] X. Zhang, B. Ni, Y. Yang, L. Zhang, Madapter: A better interaction between image and language for medical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 425–434.

[164] S. Ouyang, J. Zhang, X. Lin, X. Wang, Q. Chen, Y.-W. Chen, L. Lin, Lsms: Language-guided scale-aware medsegmentor for medical image referring segmentation, arXiv preprint arXiv:2408.17347 (2024).

[165] C.-M. Feng, Enhancing label-efficient medical image segmentation with text-guided diffusion models, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 253–262.

[166] W. Li, X. Xiong, P. Xia, L. Ju, Z. Ge, Tp-drseg: improving diabetic retinopathy lesion segmentation with explicit text-prompts assisted sam, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 743–753.

[167] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, et al., Segment anything, in: Proceedings of the IEEE/CVF international conference on computer vision, 2023, pp. 4015–4026.

[168] Q. En, Y. Guo, Akgnet: Attribute knowledge guided unsupervised lung-infected area segmentation, in: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, Springer, 2024, pp. 267–283.

[169] Y. W. Kim, H. Cho, S.-J. Ko, S.-W. Jung, Text knowledge-guided segment anything model for medical image segmentation, in: 2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), IEEE, 2024, pp. 1–4.

[170] S. Zhang, J. Zhang, Y. Xie, Y. Xia, Tpro: Text-prompting-based weakly supervised histopathology tissue segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 109–118.

[171] J. Liu, Y. Zhang, J.-N. Chen, J. Xiao, Y. Lu, B. A Landman, Y. Yuan, A. Yuille, Y. Tang, Z. Zhou, Clip-driven universal model for organ segmentation and tumor detection, in: Proceedings of the IEEE/CVF international conference on computer vision, 2023, pp. 21152–21164.

[172] J. Liu, Y. Zhang, K. Wang, M. C. Yavuz, X. Chen, Y. Yuan, H. Li, Y. Yang, A. Yuille, Y. Tang, et al., Universal and extensible language-vision models for organ segmentation and tumor detection from abdominal computed tomography, Medical image analysis 97 (2024) 103226.

[173] D. Sharma, S. Purushotham, C. K. Reddy, Medfusenet: An attention-based multimodal deep learning model for visual question answering in the medical domain, Scientific Reports 11 (1) (2021) 19826.

[174] C. Liu, S. Cheng, M. Shi, A. Shah, W. Bai, R. Arcucci, Imitate: Clinical prior guided hierarchical vision-language pre-training, IEEE Transactions on Medical Imaging (2024).

[175] Z. Cai, L. Lin, H. He, P. Cheng, X. Tang, Uni4eye++: A general masked image modeling multi-modal pre-training framework for ophthalmic image classification and segmentation, IEEE Transactions on Medical Imaging (2024).

[176] L. Peng, S. Cai, Z. Wu, H. Shang, X. Zhu, X. Li, Mmgpl: Multimodal medical data analysis with graph prompt learning, Medical Image Analysis 97 (2024) 103225.

[177] S. Sengupta, C. Heaton, S. Ghosh, W. Yin, P. Nakov, S. Wang, Top-training: Target-oriented pretraining for medical extractive question answering, in: Proceedings of the 31st International Conference on Computational Linguistics, 2025, pp. 7035–7054.

[178] OpenAI, Gpt-4 technical report (2024). arXiv:2303.08774.
URL https://arxiv.org/abs/2303.08774

[179] S. Bae, D. Kyung, J. Ryu, E. Cho, G. Lee, S. Kweon, J. Oh, L. Ji, E. I.-C. Chang, T. Kim, E. Choi, Ehrxqa: A multi-modal question answering dataset for electronic health records with chest x-ray images (2023). arXiv:2310.18652.
URL https://arxiv.org/abs/2310.18652

[180] S. Park, S. Bae, J. Kim, T. Kim, E. Choi, Graph-text multi-modal pre-training for medical representation learning (2022). arXiv:2203.09994.
URL https://arxiv.org/abs/2203.09994

[181] J.-E. Ding, C.-C. Hsu, C.-H. Chu, S. Wang, F. Liu, Enhancing multimodal medical image classification using cross-graph modal contrastive learning (2025). arXiv:2410.17494.
URL https://arxiv.org/abs/2410.17494

[182] C. Wu, X. Zhang, Y. Zhang, Y. Wang, W. Xie, Medklip: Medical knowledge enhanced language-image pre-training in radiology (2023). arXiv:2301.02228.
URL https://arxiv.org/abs/2301.02228

[183] S. Eslami, C. Meinel, G. de Melo, PubMedCLIP: How much does CLIP benefit visual question answering in the medical domain?, in: A. Vlachos, I. Augenstein (Eds.), Findings of the Association for Computational Linguistics: EACL 2023, Association for Computational Linguistics, Dubrovnik, Croatia, 2023, pp. 1181–1193. doi:10.18653/v1/2023.findings-eacl.88.
URL https://aclanthology.org/2023.findings-eacl.88/

[184] J. Liu, Z. Wang, Q. Ye, D. Chong, P. Zhou, Y. Hua, Qilin-med-vl: Towards chinese large vision-language model for general healthcare (2023). arXiv:2310.17956.

[185] M. Moor, Q. Huang, S. Wu, M. Yasunaga, Y. Dalmia, J. Leskovec, C. Zakka, E. P. Reis, P. Rajpurkar, Med-flamingo: a multimodal medical few-shot learner, in: ML4H@NeurIPS, Vol. 225 of Proceedings of Machine Learning Research, PMLR, 2023, pp. 353–367.

[186] O. C. Thawakar, A. M. Shaker, S. S. Mullappilly, H. Cholakkal, R. M. Anwer, S. H. Khan, J. Laaksonen, F. Khan, Xraygpt: Chest radiographs summarization using large medical vision-language models, in: BioNLP@ACL, Association for Computational Linguistics, 2024, pp. 440–448.

[187] Z. Wan, C. Liu, X. Wang, C. Tao, H. Shen, Z. Peng, J. Fu, R. Arcucci, H. Yao, M. Zhang, Meit: Multi-modal electrocardiogram instruction tuning on large language models for report generation (2024). arXiv:2403.04945.

[188] X. Zhu, Y. Hu, F. Mo, M. Li, J. Wu, Uni-med: A unified medical generalist foundation model for multi-task learning via connector-moe (2024). arXiv:2409.17508.

[189] G. Sun, C. Qin, H. Fu, L. Wang, Z. Tao, Self-training large language and vision assistant for medical question answering, in: EMNLP, Association for Computational Linguistics, 2024, pp. 20052–20060.

[190] D. Hein, Z. Chen, S. Ostmeier, J. Xu, M. Varma, E. P. Reis, A. E. Michalson, C. Bluethgen, H. J. Shin, C. Langlotz, A. S. Chaudhari, Preference fine-tuning for factuality in chest x-ray interpretation models without human feedback (2024). arXiv:2410.07025.

[191] F. Wang, W. Zhou, J. Y. Huang, N. Xu, S. Zhang, H. Poon, M. Chen, mdpo: Conditional preference optimization for multimodal large language models, in: EMNLP, Association for Computational Linguistics, 2024, pp. 8078–8088.

[192] K. Zhu, P. Xia, Y. Li, H. Zhu, S. Wang, H. Yao, Mmedpo: Aligning medical vision-language models with clinical-aware multimodal preference optimization (2024). arXiv:2412.06141.

[193] T. Xie, Z. Gao, Q. Ren, H. Luo, Y. Hong, B. Dai, J. Zhou, K. Qiu, Z. Wu, C. Luo, Logic-rl: Unleashing llm reasoning with rule-based reinforcement learning (2025). arXiv:2502.14768.

[194] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, D. Amodei, Scaling laws for neural language models, arXiv preprint arXiv:2001.08361 (2020).

[195] D. Demner-Fushman, M. D. Kohli, M. B. Rosenman, S. E. Shooshan, L. Rodriguez, S. Antani, G. R. Thoma, C. J. McDonald, Preparing a collection of radiology examinations for distribution and retrieval, Journal of the American Medical Informatics Association 23 (2) (2016) 304–310.

[196] A. Bustos, A. Pertusa, J.-M. Salinas, M. De La Iglesia-Vaya, Padchest: A large chest x-ray image dataset with multi-label annotated reports, Medical image analysis 66 (2020) 101797.

[197] S. Soni, M. Gudala, A. Pajouhi, K. Roberts, Radqa: A question answering dataset to improve comprehension of radiology reports, in: Proceedings of the thirteenth language resources and evaluation conference, 2022, pp. 6250–6259.

[198] A. Ben Abacha, S. A. Hasan, V. V. Datla, D. Demner-Fushman, H. Müller, Vqa-med: Overview of the medical visual question answering task at imageclef 2019, in: Proceedings of CLEF (Conference and Labs of the Evaluation Forum) 2019 Working Notes, 9-12 September 2019, 2019.

[199] J. J. Lau, S. Gayen, A. Ben Abacha, D. Demner-Fushman, A dataset of clinically generated visual questions and answers about radiology images, Scientific data 5 (1) (2018) 1–10.

[200] J. Chen, C. Gui, R. Ouyang, A. Gao, S. Chen, G. H. Chen, X. Wang, R. Zhang, Z. Cai, K. Ji,

et al., Huatuogpt-vision, towards injecting medical visual knowledge into multimodal llms at scale, arXiv preprint arXiv:2406.19280 (2024).

[201] X. He, Y. Zhang, L. Mou, E. Xing, P. Xie, Pathvqa: 30000+ questions for medical visual question answering, arXiv preprint arXiv:2003.10286 (2020).

[202] Y. Xie, C. Zhou, L. Gao, J. Wu, X. Li, H.-Y. Zhou, S. Liu, L. Xing, J. Zou, C. Xie, et al., Medtrinity-25m: A large-scale multimodal dataset with multigranular annotations for medicine, arXiv preprint arXiv:2408.02900 (2024).

[203] S. P. Morozov, A. E. Andreychenko, N. A. Pavlov, A. Vladzymyrskyy, N. V. Ledikhova, V. A. Gombolevskiy, I. A. Blokhin, P. B. Gelezhe, A. Gonchar, V. Y. Chernina, Mosmeddata: Chest ct scans with covid-19 related findings dataset, arXiv preprint arXiv:2005.06465 (2020).

[204] J. Hofmanninger, F. Prayer, J. Pan, S. Röhrich, H. Prosch, G. Langs, Automatic lung segmentation in routine imaging is primarily a data diversity problem, not a methodology problem, European radiology experimental 4 (2020) 1–13.

[205] A. Degerli, S. Kiranyaz, M. E. Chowdhury, M. Gabbouj, Osegnet: Operational segmentation network for covid-19 detection using chest x-ray images, in: 2022 IEEE International Conference on Image Processing (ICIP), IEEE, 2022, pp. 2306–2310.

[206] O. Pelka, S. Koitka, J. Rückert, F. Nensa, C. M. Friedrich, Radiology objects in context (roco) (2018).

[207] S. Subramanian, L. L. Wang, S. Mehta, B. Bogin, M. van Zuylen, S. Parasa, S. Singh, M. Gardner, H. Hajishirzi, Medicat: A dataset of medical images, captions, and textual references, arXiv preprint arXiv:2010.06000 (2020).

[208] W. Lin, Z. Zhao, X. Zhang, C. Wu, Y. Zhang, Y. Wang, W. Xie, Pmc-clip: Contrastive language-image pre-training using biomedical documents, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2023, pp. 525–536.

[209] C.-Y. Lin, Rouge: A package for automatic evaluation of summaries, in: Text summarization branches out, 2004, pp. 74–81.

[210] K. Papineni, S. Roukos, T. Ward, W.-J. Zhu, Bleu: a method for automatic evaluation of machine translation, in: Proceedings of the 40th annual meeting of the Association for Computational Linguistics, 2002, pp. 311–318.

[211] S. Banerjee, A. Lavie, Meteor: An automatic metric for mt evaluation with improved correlation with human judgments, in: Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization, 2005, pp. 65–72.

[212] R. Vedantam, C. Lawrence Zitnick, D. Parikh, Cider: Consensus-based image description evaluation, in: Proceedings of the IEEE conference on computer vision and pattern recognition, 2015, pp. 4566–4575.

[213] P. Kumar, S. K. Sahu, Sim-bert: Speech intelligence model using nlp-bert with improved accuracy, in: Artificial Intelligence and Speech Technology, CRC Press, 2021, pp. 439–445.

[214] J. Bertels, T. Eelbode, M. Berman, D. Vandermeulen, F. Maes, R. Bisschops, M. B. Blaschko, Optimizing the dice score and jaccard index for medical image segmentation: Theory and practice, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part II 22, Springer, 2019, pp. 92–100.

[215] Z. Li, L. Yu, J. Zhang, Distribution-aware margin calibration for medical image segmentation, arXiv preprint arXiv:2011.01462 (2020).

[216] J. Delbrouck, P. J. Chambon, C. Bluethgen, E. B. Tsai, O. Almusa, C. P. Langlotz, Improving the factual correctness of radiology report generation with semantic rewards, in: EMNLP (Findings), Association for Computational Linguistics, 2022, pp. 4348–4360.

[217] J. Wang, Y. Huang, Z. Liu, D. Xu, C. Wang, X. Shi, R. Guan, H. Wang, W. Yue, Y. Huang, Stampsy: Towards spatiotemporal-aware mixed-type dialogues for psychological counseling, arXiv preprint arXiv:2412.16674 (2024).

[218] L. Liu, X. Yang, J. Lei, Y. Shen, J. Wang, P. Wei, Z. Chu, Z. Qin, K. Ren, A survey on medical large language models: Technology, application, trustworthiness, and future directions (2024). arXiv:2406.03712.

[219] Z. Su, J. Zhang, X. Qu, T. Zhu, Y. Li, J. Sun, J. Li, M. Zhang, Y. Cheng, Conflictbank: A benchmark for evaluating the influence of knowledge conflicts in llms, in: NeurIPS, 2024.

[220] J. Wu, Y. Kim, H. Wu, Hallucination benchmark in medical visual question answering, in: Tiny Papers @ ICLR, OpenReview.net, 2024.

[221] P. Xia, Z. Chen, J. Tian, Y. Gong, R. Hou, Y. Xu, Z. Wu, Z. Fan, Y. Zhou, K. Zhu, W. Zheng, Z. Wang, X. Wang, X. Zhang, C. Bansal, M. Niethammer, J. Huang, H. Zhu, Y. Li, J. Sun, Z. Ge, G. Li, J. Y. Zou, H. Yao, CARES: A comprehensive benchmark of trustworthiness in medical vision language models, in: NeurIPS, 2024.

[222] C. Royer, B. Menze, A. Sekuboyina, Multimedeval: A benchmark and a toolkit for evaluating medical vision-language models (2024). arXiv:2402.09262.

[223] Y. Zhou, T. L. H. Faith, Y. Xu, S. Leng, X. Xu, Y. Liu, R. S. M. Goh, Benchx: A unified benchmark framework for medical vision-language pretraining on chest x-rays, in: NeurIPS, 2024.

[224] X. Hong, Z. Song, L. Li, X. Wang, F. Liu, BESTMVQA: A benchmark evaluation system for medical visual question answering, in: ECML/PKDD (9), Vol. 14949 of Lecture Notes in Computer Science, Springer, 2024, pp. 435–451.

[225] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, et al., A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions, ACM Transactions on Information Systems 43 (2) (2025) 1–55.

[226] V. Arsenyan, S. Bughdaryan, F. Shaya, K. W. Small, D. Shahnazaryan, Large language models for biomedical knowledge graph construction: Information extraction from emr notes, in: Proceedings of the 23rd Workshop on Biomedical Natural Language Processing, 2024, pp. 295–317.

[227] Y. Luo, Z. Liu, L. Wang, B. Wu, J. Zheng, Q. Ma, Knowledge-empowered dynamic graph network for irregularly sampled medical time series, Advances in Neural Information Processing Systems 37 (2025) 67172–67199.

[228] W. Liang, P. D. Meo, Y. Tang, J. Zhu, A survey of multimodal knowledge graphs: Technologies and trends, ACM Computing Surveys 56 (11) (2024) 1–41.

[229] Z. Wang, Z. Wang, B. Srinivasan, V. N. Ioannidis, H. Rangwala, R. ANUBHAI, Biobridge: Bridging biomedical foundation models via knowledge graphs, in: The Twelfth International Conference on Learning Representations.

[230] Q. Sun, A. Akman, B. W. Schuller, Explainable artificial intelligence for medical applications: A review, ACM Transactions on Computing for Healthcare 6 (2) (2025) 1–31.

[231] M. I. Hossain, G. Zamzmi, P. R. Mouton, M. S. Salekin, Y. Sun, D. Goldgof, Explainable ai for medical data: current methods, limitations, and future directions, ACM Computing Surveys 57 (6) (2025) 1–46.

[232] H. Rasheed, M. Maaz, S. Shaji, A. Shaker, S. Khan, H. Cholakkal, R. M. Anwer, E. Xing, M.-H. Yang, F. S. Khan, Glamm: Pixel grounding large multimodal model, The IEEE/CVF Conference on Computer Vision and Pattern Recognition (2024).

[233] S. Zuo, Q. Zhang, C. Liang, P. He, T. Zhao, W. Chen, Moebert: from bert to mixture-of-experts via importance-guided adaptation, in: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2022.

[234] B. Lin, Z. Tang, Y. Ye, J. Huang, J. Zhang, Y. Pang, P. Jin, M. Ning, J. Luo, L. Yuan, Moe-llava: Mixture of experts for large vision-language models (2024). arXiv:2401.15947.

[235] N. Ibrahim, S. AboulEla, A. F. Ibrahim, R. F. Kashef, A survey on augmenting knowledge graphs (kgs) with large language models (llms): models, evaluation metrics, benchmarks, and challenges, Discov. Artif. Intell. 4 (1) (2024) 76.

[236] X. Wang, G. Cheng, A survey on extractive knowledge graph summarization: Applications, approaches, evaluation, and future directions, in: IJCAI, ijcai.org, 2024, pp. 8290–8298.

[237] Y. Jiang, Y. Wang, C. Wu, W. Zhong, X. Zeng, J. Gao, L. Li, X. Jiang, L. Shang, R. Tang, et al., Learning to edit: Aligning llms with knowledge editing, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 4689–4705.

[238] K. Pan, Z. Fan, J. Li, Q. Yu, H. Fei, S. Tang, R. Hong, H. Zhang, Q. Sun, Towards unified multimodal editing with enhanced knowledge collaboration, Advances in Neural Information Processing Systems 37 (2025) 110290–110314.

[239] E. J. Hu, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al., Lora: Low-rank adaptation of large language models, in: International Conference on Learning Representations.

[240] M. Jovanovic, P. Voss, Towards incremental learning in large language models: A critical review, arXiv preprint arXiv:2404.18311 (2024).

[241] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, Y. Zhang, A survey on large language model (llm) security and privacy: The good, the bad, and the ugly, High-Confidence Computing (2024) 100211.

[242] J. Zhang, K. Qian, H. Luo, Y. Liu, X. Qiao, X. Xu, J. Tian, Process monitoring for tower pumping units under variable operational conditions: From an integrated multitasking perspective, Control Engineering Practice 156 (2025) 106229.

[243] J. Zhang, J. Tian, A. M. Alcaide, J. I. Leon, S. Vazquez, L. G. Franquelo, H. Luo, S. Yin, Lifetime extension approach based on the levenberg–marquardt neural network and power routing of dc–dc converters, IEEE Transactions on Power Electronics 38 (8) (2023) 10280–10291.

[244] A. Plaat, A. Wong, S. Verberne, J. Broekens, N. van Stein, T. Back, Reasoning with large language models, a survey, arXiv preprint arXiv:2407.11511 (2024).